

A Mini Project Report on
**EXPERT SYSTEM APPLICATION DESIGN FOR
ENHANCED DENTAL DISEASE DIAGNOSIS**

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JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY, HYDERABAD

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BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

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2020- 2024

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CERTIFICATE

This is to certify that the project work entitled **“EXPERT SYSTEM APPLICATION DESIGN FOR ENHANCED DENTAL DISEASE DIAGNOSIS”** is submitted by Mr Sanem.Sree Malik Goud, Mr S. Sai Kiran, Ms N. Swetha bonafied students of **Kommuri Pratap Reddy Institution of Technology** in partial fulfilment of the requirement for the reward of the Degree of Bachelor of Technology in **Computer Science and Engineering** of the **Jawaharlal Nehru Technological University Hyderabad**, during the year 2023-24.

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DECLARATION

We hereby declare that this project work entitled “**EXPERT SYSTEM APPLICATION DESIGN FOR ENHANCED DENTAL DISEASE DIAGNOSIS**” in partial fulfilment of requirements for the award of degree of **Computer Science and Engineering** is a bonafide work carried out by us during the academic year 2023- 24.

We further declare that this project is a result of our effort and has not been submitted for the award of any degree by us to any institute.

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KOMMURI PRATAP REDDY INSTITUTE OF TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Vision of the Institute

To emerge as a premier institute for high quality professional graduates who can contribute to economic and social developments of the Nation.

Mission of the Institute

Mission	Statement
IM₁	To have holistic approach in curriculum and pedagogy through industry interface to meet the needs of Global Competency.
IM₂	To develop students with knowledge, attitude, employability skills, entrepreneurship, research potential and professionally Ethical citizens.
IM₃	To contribute to advancement of Engineering & Technology that would help to satisfy the societal needs.
IM₄	To preserve, promote cultural heritage, humanistic values and Spiritual values thus helping in peace and harmony in the society.



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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

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To Provide Quality Education in Computer Science for the innovative professionals to work for the development of the nation.

Mission of the Department

Mission	Statement
DM1	Laying the path for rich skills in Computer Science through the basic knowledge of mathematics and fundamentals of engineering
DM2	Provide latest tools and technology to the students as a part of learning Infrastructure
DM3	Training the students towards employability and entrepreneurship to meet the societal needs.
DM4	Grooming the students with professional and social ethics

Program Educational Objectives (PEOs)

<i>PEO'S</i>	Statement
PEO1	The graduates of Computer Science and Engineering will have successful career in technology.
PEO2	The graduates of the program will have solid technical and professional foundation to continue higher studies.
PEO3	The graduate of the program will have skills to develop products, offer services and innovation.
PEO4	The graduates of the program will have fundamental awareness of industry process, tools and technologies.

Program Outcomes

PO1	Engineering Knowledge: Apply the knowledge of mathematics, science, Engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem Analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	Design/development of Solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO7	Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental context, and demonstrate the knowledge of, and need for sustainable development.

PO8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
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PO9	Individual and team network: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, being able to comprehend and write effective reports and design documentation, make Effective presentations, and give and receive clear instructions.
PO11	Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environment.
PO12	Life-Long learning: Recognize the need for, and have the preparation and able to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES

PSO1	Foundation of mathematical concepts: To use mathematical methodologies to crack problem using suitable mathematical analysis, data structure and suitable algorithm.
PSO2	Foundation of Computer Science: The ability to interpret the fundamental concepts and methodology of computer systems. Students can understand the functionality of hardware and software aspects of computer systems.
PSO3	Foundation of Software development: The ability to grasp the software development lifecycle and methodologies of software systems. Possess competent skills and knowledge of software design process.

ABSTRACT

Dentistry is an ever-evolving field, and nowadays, there's a strong emphasis on harnessing technology to improve the diagnosis and treatment of dental diseases. We all know how common issues like cavities, gum disease, and oral infections can lead to serious oral health problems if not dealt with properly. In the past, dentists relied solely on their skills and experience to diagnose these conditions. They would manually examine patients, study the data, and take into consideration the patient's medical history to arrive at a diagnosis. While this approach has served dentistry well for many years, it does come with its limitations. One of the drawbacks is the possibility of human errors and biases affecting the accuracy of diagnoses. Additionally, the time-consuming process of manually analyzing large datasets slows down the diagnosis. What's more, the traditional approach lacks the ability to integrate real-time updates from the latest dental research and advancements in dental care. As a result, there's a growing demand for a smarter system that can aid dentists in making quicker and more precise diagnoses. To address these challenges and improve the accuracy and efficiency of dental disease diagnosis, experts have been exploring the potential of expert systems, a type of artificial intelligence (AI) that mimics human decision-making abilities. These systems have the capacity to revolutionize dental diagnosis by providing timely and accurate assessments, leading to more effective treatment plans and better patient outcomes. By combining the expertise of human dentists with the capabilities of AI, this system can enhance the overall performance of dental professionals, leading to better patient outcomes and improved oral health. Furthermore, the development of this application highlights the promising role of AI in revolutionizing healthcare practices, setting the stage for even more advancements in medical diagnostics.

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CHAPTER 1

1.INTRODUCTION

1.1 Overview

Designing an expert system for enhanced dental disease diagnosis involves the creation of a specialized software application that leverages artificial intelligence and domain-specific knowledge to assist dental professionals in accurately diagnosing and treating oral health issues. This system aims to streamline the diagnostic process, reduce human errors, and improve the overall quality of dental care.

The core of this expert system is its knowledge base, which contains a vast repository of dental information, including symptoms, causes, risk factors, and treatment options for various dental diseases and conditions. This knowledge is structured in a way that mirrors the decision-making process of a skilled dentist. To develop the knowledge base, a team of experts in dentistry, along with AI specialists, collaborate to ensure that the system's information is up-to-date and comprehensive.

The system also incorporates a rule-based inference engine that uses logical rules and algorithms to analyze patient data. Dentists or dental hygienists input patient information such as medical history, symptoms, and diagnostic test results into the system. The expert system then applies its knowledge to suggest potential diagnoses and treatment recommendations. It can also provide a list of differential diagnoses, helping practitioners consider various possibilities.

User interaction is a crucial aspect of the application design. The system should have a user-friendly interface that allows dental professionals to input patient data easily and receive clear and concise diagnostic recommendations. It can also provide educational materials and explanations to help clinicians understand the reasoning behind the system's suggestions.

Furthermore, to enhance its capabilities, the expert system can be integrated with imaging technology like X-rays and intraoral cameras. By analyzing these visual data, it can assist in identifying dental conditions that are not immediately apparent through patient-reported symptoms alone.

To ensure the accuracy and reliability of the expert system, it should undergo rigorous testing and validation with real patient cases, and continuous updates should be made to the knowledge base as new research and treatment methods emerge.

1.2 Research Motivation

The motivation for developing an expert system for enhanced dental disease diagnosis stems from several critical factors within the field of dentistry and healthcare in general.

First and foremost, dental health is a fundamental aspect of overall well-being, with oral diseases having a substantial impact on individuals' quality of life. Timely and accurate diagnosis is crucial in preventing the progression of oral conditions, which can range from common issues like cavities and gum disease to more severe problems such as oral cancer. An expert system can significantly contribute to early detection and appropriate treatment, ultimately improving patients' oral health outcomes.

Additionally, there is a growing demand for more efficient and effective healthcare delivery systems. Dentists and dental professionals often face time constraints and increasing patient loads. The automation and support provided by an expert system can help streamline the diagnostic process, allowing practitioners to make better use of their time and expertise. This, in turn, can lead to quicker assessments, reduced waiting times for patients, and improved access to dental care, especially in underserved areas.

Furthermore, dental practice, like many other medical disciplines, benefits from advances in technology and artificial intelligence. The integration of AI into dentistry can enhance the precision of diagnoses by considering a broader range of data and offering intelligent suggestions based on vast amounts of clinical knowledge. This not only aids in more accurate diagnoses but also supports dental professionals in making well-informed decisions about treatment plans and referrals.

The expert system can also contribute to ongoing dental education and training. Dental practitioners can use it as a valuable tool for continuous learning, staying updated on the latest research findings, treatment methods, and best practices in the field. This educational aspect can further elevate the quality of dental care provided to patients.

Lastly, from a research perspective, the development of such a system offers opportunities to gather valuable data on dental disease patterns, treatment effectiveness, and patient outcomes. Researchers can use this information to refine diagnostic algorithms and treatment recommendations, contributing to the overall improvement of dental healthcare.

1.3 Problem Statement

The problem statement for the development of an expert system for enhanced dental disease diagnosis can be articulated as follows:

In contemporary healthcare, particularly within the field of dentistry, there exists a pressing challenge related to the accurate and timely diagnosis of dental diseases and conditions. Dental health is an integral component of overall well-being, with oral ailments significantly impacting individuals' quality of life. However, the current diagnostic processes in dental practice often rely heavily on the expertise and experience of dental professionals, which can lead to variations in diagnosis, delayed treatments, and, in some cases, misdiagnosis. Moreover, the increasing patient loads and time constraints faced by dental practitioners can further exacerbate these challenges, potentially compromising the quality of care provided.

This problem is exacerbated by the constant evolution of dental knowledge, including new research findings, emerging treatment modalities, and evolving diagnostic techniques. Dental professionals are often inundated with an overwhelming volume of information that they must continually assimilate to stay current in their practice. Additionally, the lack of access to specialized dental care in remote or underserved areas exacerbates the issue of timely diagnosis and appropriate treatment.

Furthermore, the integration of advanced technology and artificial intelligence (AI) into dentistry presents an opportunity to address these challenges effectively. However, the development and implementation of AI-driven expert systems for dental diagnosis are still in their infancy, with limited adoption and validation in real-world dental practice settings. This gap highlights the need for a comprehensive and well-designed expert system that can harness the power of AI and domain-specific knowledge to provide accurate, efficient, and consistent dental disease diagnosis.

Therefore, the problem at hand is twofold: the need to enhance the accuracy and efficiency of dental disease diagnosis to improve patient outcomes and the requirement to bridge the gap

between evolving dental knowledge and its practical application in clinical settings. Addressing these issues through the development of a robust expert system tailored to the domain of dentistry represents a vital step toward addressing the existing challenges in dental healthcare and ensuring that patients receive the highest standard of care irrespective of geographic location or the expertise of their dental practitioner.

1.4 Applications

The development of an expert system for enhanced dental disease diagnosis holds vast potential across various applications within the field of dentistry and healthcare as a whole.

- **Clinical Diagnosis and Treatment Planning:** The primary application lies in clinical practice, where dental professionals can use the expert system as a valuable diagnostic tool. It can assist in accurately identifying dental diseases and conditions, offering recommendations for treatment plans, and providing insights into the best practices for patient care. This can result in more precise and evidence-based treatments, reducing the risk of misdiagnosis and ensuring optimal patient outcomes.
- **Remote and Underserved Areas:** In regions with limited access to specialized dental care, the expert system can be a game-changer. Dental practitioners in these areas can rely on the system's guidance to make informed diagnoses and treatment decisions, bridging the gap in quality dental healthcare between urban and rural communities.
- **Dental Education and Training:** The expert system can serve as a powerful educational tool for dental students, hygienists, and practitioners. It can facilitate hands-on learning by providing real-world case scenarios, diagnostic challenges, and treatment recommendations. This application can enhance the quality of dental education and ensure that emerging dental professionals are well-prepared and up-to-date with the latest advancements in the field.
- **Research and Data Collection:** The system can contribute to dental research by collecting and anonymizing patient data. Researchers can analyze this data to gain insights into disease patterns, treatment effectiveness, and emerging trends in dental health. This can lead to the development of new treatment modalities and the refinement of existing diagnostic algorithms.
- **Continuous Quality Improvement:** Dental practices can use the expert system to assess and improve the quality of their services. By analyzing the system's recommendations and

comparing them to clinical outcomes, dental professionals can identify areas for improvement and implement evidence-based protocols for enhanced patient care.

- **Teledentistry:** In the era of telehealth, the expert system can be integrated into teledentistry platforms, allowing remote consultations with dental specialists. This can expand access to dental expertise, particularly for patients who cannot easily visit a dental clinic in person.
- **Efficiency and Workflow Enhancement:** Dental clinics can benefit from the system's ability to streamline workflows. It can assist in patient triage, prioritizing cases based on severity, and facilitating efficient scheduling for treatments and follow-up appointments. This can optimize resource allocation and reduce wait times for patients.
- **Patient Empowerment:** The expert system can also empower patients by providing them with information about their dental health and treatment options. Patients can better understand their conditions and treatment plans, making them active participants in their oral healthcare decisions.

CHAPTER 2

2.LITERATURE SURVEY

Over the past decade, artificial intelligence (AI) has made remarkable contributions to various subdisciplines falling under the category of dentistry, specifically periodontology. Different studies have explored dental disease detection, localization, classification, and segmentation within the dental domain (e.g., [1]). However, few studies have explored dental disease localization as discussed in the literature. From the existing literature, several challenges are found regarding dental carious region localization. A comprehensive overview of existing studies is presented in Table 1. Further exploration is required to propose detection and localization approaches for dental caries diagnosis in real time.

To classify enamel, dentin, and pulp caries, Oprea et al., proposed rule-based classification. The authors were able to categorize regions as dentin caries sized over 2 mm [9]. Another rule-based approach based on the gradient histogram and threshold was proposed by ALbahbah and fellow authors on panoramic radiographs to extract and segment decayed and normal teeth [10]. Lin et al., investigated the level segmentation method based using SVM (support vector machine), KNN (K nearest neighbor), and a Bayesian classifier for localizing alveolar bone loss [11]. Results show that the model can localize alveolar bone loss with higher classification accuracy. A cluster-based segmentation technique was proposed by Datta and Chaki to detect dental cavities in [12]. The proposed model utilized a Wiener filter to extract caries lesions followed by region segmentation to monitor the lesion size and growth. To detect and classify proximal carious and non-carious lesions on panoramic radiographs, Na'am et al., explored multiple morphological gradient-based image processing methods on images with manually cropped regions [13].

Different deep learning approaches have been employed by researchers to pave way for more efficient and effective methods to diagnose dental caries. To classify carious and non-carious teeth on a small labeled dataset, a pre-trained CNN was utilized by Prajapati et al. [14]. The model was able to classify dental caries, periodontitis, and periapical infection. Lee et al. utilized a deep CNN to diagnose and classify caries using 3000 periapical radiographs [15]. The model achieved an AUC of 0.91 for premolar, 0.89 for molar, and 0.84 for both premolar and molar models. For the identification of dental caries, Cantu et al., investigated U-Net on bitewing radiographs [16]. It

was found that segmentation-based models possess the potential to aid dental clinicians in detecting and locating dental caries more efficiently. For the identification of endo-perio lesions on periapical radiographs, Sajjad et al., investigated AlexNet, for which the model achieved an accuracy of 98% [17]. For early identification of dental caries, Kumari et al., preprocessed bitewing radiographic images using contrast limited adaptive histogram equalization (CLAHE) and noise filtering followed by a meta-heuristic based ResNeXt RNN (recurrent neural network) [18].

Radiological examinations help dental clinicians in the identification of teeth abnormalities, cysts, infections, and infections. However, manual examinations are time-consuming and rely solely on a specialist's opinion which may bring differences in the diagnosis. Different methods have been employed by researchers in recent years mainly relying on boundary-based, region-based [19], cluster-based, and threshold-based methods [11]. As the first step, Jader et al., employed an RCNN for the segmentation of caries and the detection of missing teeth on buccal images. The results indicated that deep learning-based instance segmentation has the potential to automate the process of caries detection and medical report generation [2].

The faster region-based convolutional neural network (Faster-RCNN), which extends the Fast-RCNN is utilized to localize teeth lesions [5]. The model achieves both a recall and precision of above 90%, however, the model suffers in numbering the teeth in complicated cases. A Faster-RCNN built on the region proposal network (RPN) and object detection network (ODN) detected different types of teeth achieving a mean average precision (mAP) of 91.40% and an accuracy of 91.03% [6]. However, the model was applied to a small dataset and performance can not be generalized. Another variant of Faster-RCNN pre-trained on ResNet-50 was employed in [7] for the detection of carious teeth, achieving a precision of 73.49% and an F1 score of 0.68. The model, however, does not identify the type of caries and only localizes the caries region.

An M-RCNN, which extends the Faster-RCNN with pre-trained ResNet-101 was found to be helpful in the identification of missing or broken teeth, achieving an accuracy of 98% [2]. However, segmentation performance metrics were not reported in the study. For pixel-wise segmentation of visible light images for identification of oral cavities [3], M-RCNN achieves an accuracy of 74.4%. However, the dataset is sparse and other relevant performance metrics have not been reported for comparison. In another attempt, an M-RCNN with a fully convolutional network (FCN) and a ResNet-101 backbone [4] was investigated to localize occlusal surface caries

on a limited dataset, but the computational complexity was not reported. In a recent attempt, a hybrid M-RCNN [8] was employed to identify dental caries on mixed images achieving an average precision of 81.02% and an accuracy of 95.75%, however, the model does not identify caries type for both colored and X-ray images. Additionally, an M-RCNN with ResNet as its backbone requires a substantial amount of calculations to learn and analyze, and the training process for M-RCNN requires high-performance computational resources such as GPU and memory [20].

Table 1. Strengths and weaknesses of baseline dental lesion localization models.

Study	Image Modality	Task	Method	Strengths	Weaknesses
Jader et al., 2018 [2]	Panoramic images	Localize missing teeth	M-RCNN with ResNet-101 backbone	The model is helpful in identifying missing or broken teeth with an accuracy of 98%	- Highly variable data - Other metrics i.e., mAP, IoU are not reported for comparison
Anantharaman et al., 2018 [3]	Colored images	Detect and segment cold/canker sores	M-RCNN with ResNet-101 backbone	The model is helpful in performing pixel-wise segmentation of visible light images of oral cavity with accuracy of 74.4%	- Sparse dataset - Other metrics i.e., mAP, IoU, precision, F1-score, recall are not reported for comparison
Moutselos et al., 2019 [4]	Colored images	Localize and classify caries occlusal surfaces	M-RCNN with FCN and ResNet-101 backbone	The model provided encouraging performance for automatically selecting image texture features and detect lesions without additional pre-processing actions	- The computational complexity is not reported
Chen et al., 2019 [5]	Periapical radiographs	Teeth localization	Faster-RCNN	The model detects and numbers teeth with recall and	- The model suffers in numbering teeth in

		and numbering		precision exceeding 90% on manually annotated dataset	complicated cases such as heavily decayed teeth
Laishram & Thongam, 2020 [6]	Panoramic radiographs	Localize and classify different type of teeth	Faster-RCNN built on RPN and ODN	The model is helpful in detecting different types of teeth achieving mean average precision (mAP) of 91.40% and accuracy of 91.03%	- Limited dataset in terms of size
Zhu et al., 2022 [7]	Periapical radiographs	Detection of carious teeth	Faster-RCNN with pretrained ResNet-50	The model is helpful in with an average precision of 73.49%, F1-score of 0.68 with sample detection speed of 0.1923	- It suffers from computational compexity - The model does not identify caries type
Rashid et al., 2022 [8]	Mixed images (colored and periapical radiographic images)	Detect and localize dental carious regions	Hybrid M-RCNN	The model was helpful in localizing dental carious regions with a precision of 81.02% and accuracy of 95.75%	- Limited dataset in terms of size - The model does not identify caries type for both colored and X-ray image

CHAPTER 3

3.EXISTING SYSTEM

3.1 SVM

SVM is one of the most popular Supervised Learning algorithms, which is used for Classification as well as Regression problems. However, primarily, it is used for Classification problems in Machine Learning. The goal of the SVM algorithm is to create the best line or decision boundary that can segregate n-dimensional space into classes so that we can easily put the new data point in the correct category in the future. This best decision boundary is called a hyperplane.

SVM chooses the extreme points/vectors that help in creating the hyperplane. These extreme cases are called as support vectors, and hence algorithm is termed as Support Vector Machine. Consider the below diagram in which there are two different categories that are classified using a decision boundary or hyperplane:

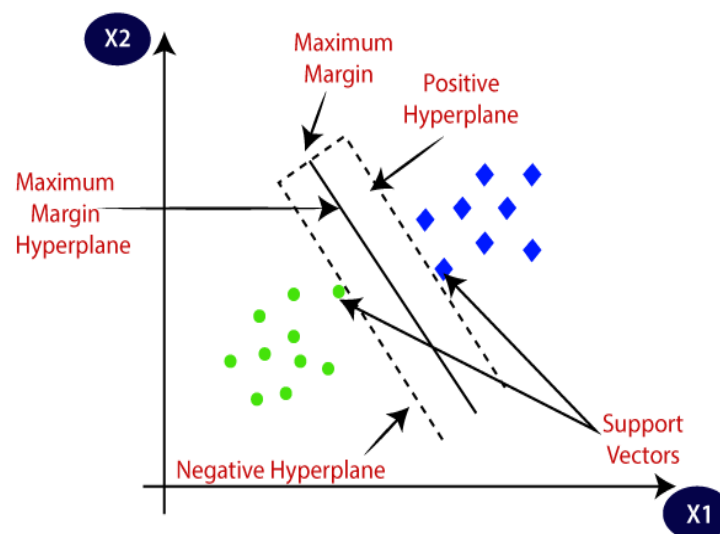


Figure 3.2 Analysis of SVM

Example: SVM can be understood with the example that we have used in the KNN classifier. Suppose we see a strange cat that also has some features of dogs, so if we want a

model that can accurately identify whether it is a cat or dog, so such a model can be created by using the SVM algorithm. We will first train our model with lots of images of cats and dogs so that it can learn about different features of cats and dogs, and then we test it with this strange creature. So as support vector creates a decision boundary between these two data (cat and dog) and choose extreme cases (support vectors), it will see the extreme case of cat and dog. On the basis of the support vectors, it will classify it as a cat. Consider the below diagram:

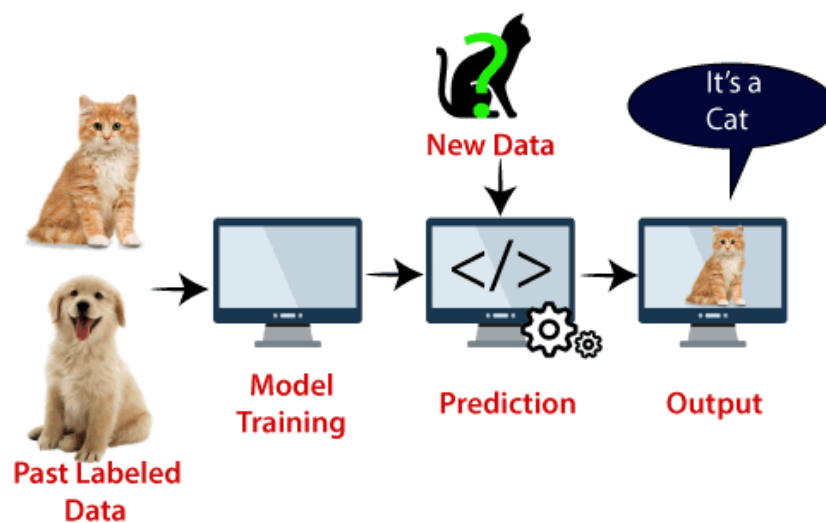


Figure 3. Basic classification using SVM

Types of SVM: SVM can be of two types:

Linear SVM: Linear SVM is used for linearly separable data, which means if a dataset can be classified into two classes by using a single straight line, then such data is termed as linearly separable data, and classifier is used called as Linear SVM classifier.

Non-linear SVM: Non-Linear SVM is used for non-linearly separated data, which means if a dataset cannot be classified by using a straight line, then such data is termed as non-linear data and classifier used is called as Non-linear SVM classifier

3.2 SVM working

Linear SVM: The working of the SVM algorithm can be understood by using an example. Suppose we have a dataset that has two tags (green and blue), and the dataset has two features x_1 and x_2 . We want a classifier that can classify the pair (x_1, x_2) of coordinates in either green or blue. Consider the below image:

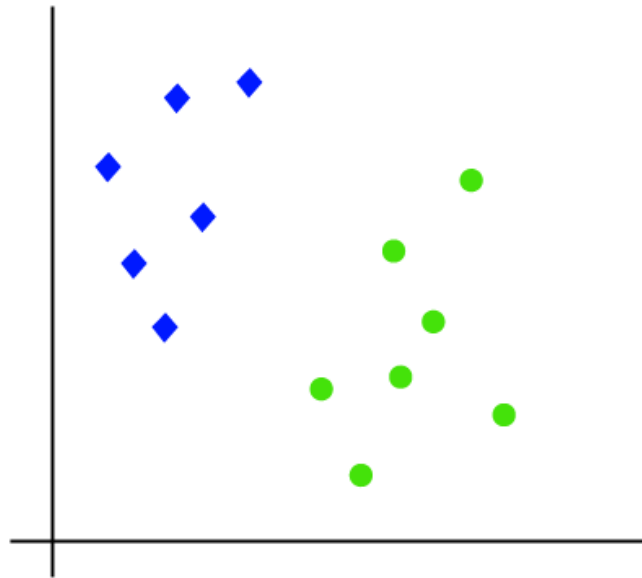


Figure 3.4. Linear SVM

So as it is 2-d space so by just using a straight line, we can easily separate these two classes. But there can be multiple lines that can separate these classes. Consider the below image:

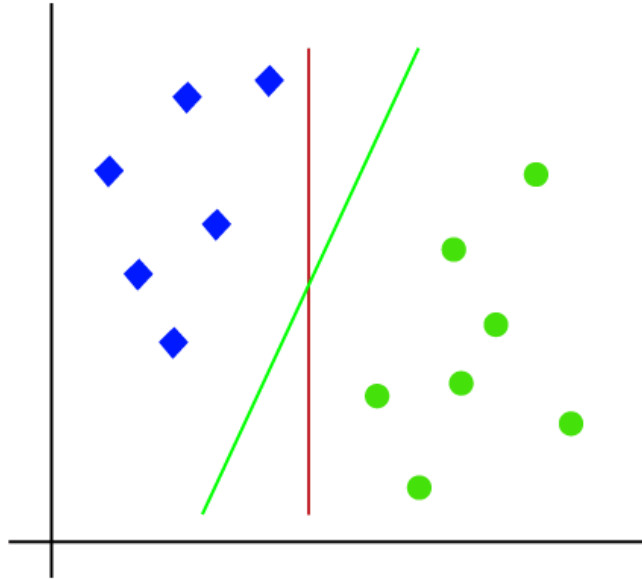


Figure 3.5. Test-Vector in SVM

Hence, the SVM algorithm helps to find the best line or decision boundary; this best boundary or region is called as a hyperplane. SVM algorithm finds the closest point of the lines from both the classes. These points are called support vectors. The distance between the vectors and the hyperplane is called as margin. And the goal of SVM is to maximize this margin. The hyperplane with maximum margin is called the optimal hyperplane.

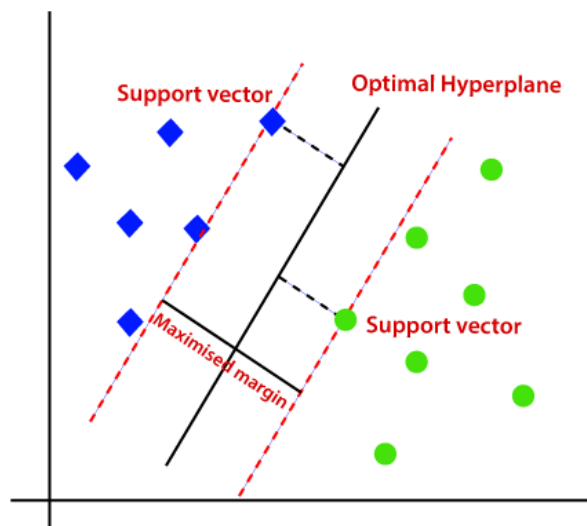


Figure 3.5. Classification in SVM

Non-Linear SVM: If data is linearly arranged, then we can separate it by using a straight line, but for non-linear data, we cannot draw a single straight line. Consider the below image:

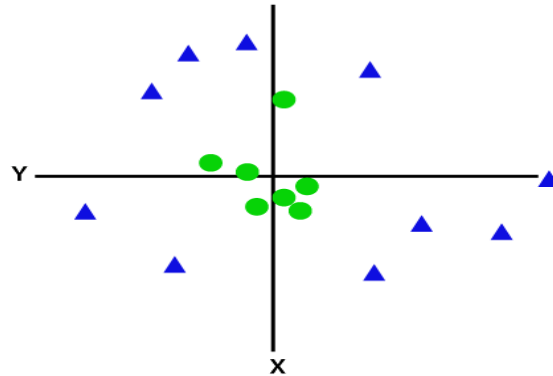


Figure 3.6. Non-Linear SVM

So, to separate these data points, we need to add one more dimension. For linear data, we have used two dimensions x and y, so for non-linear data, we will add a third-dimension z. It can be calculated as:

$$z=x^2+y^2$$

By adding the third dimension, the sample space will become as below image:

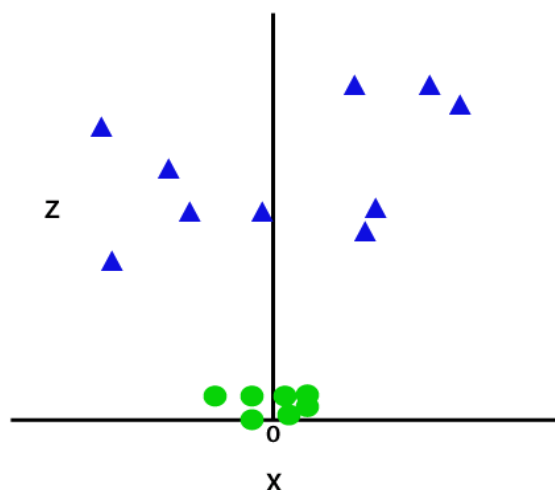


Figure 3.7. Non-Linear SVM data separation

So now, SVM will divide the datasets into classes in the following way. Consider the below image:

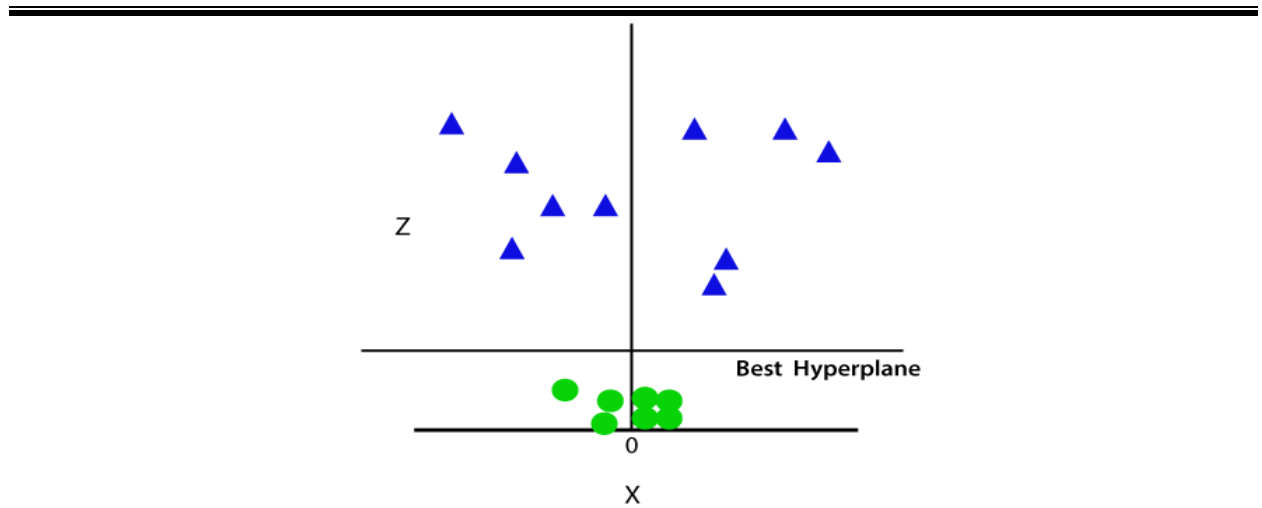


Figure 3.7. Non-Linear SVM best hyperplane

Since we are in 3-d Space, hence it is looking like a plane parallel to the x-axis. If we convert it in 2d space with $z=1$, then it will become as:

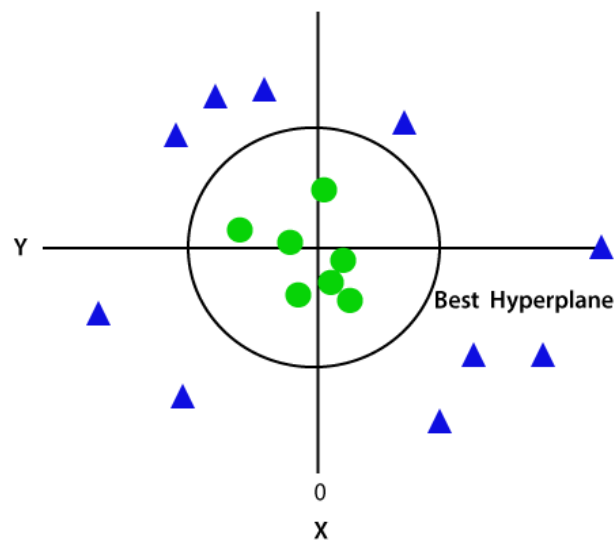


Figure 3.7. Non-Linear SVM with ROC

Hence, we get a circumference of radius 1 in case of non-linear data.

3.3 Drawbacks

Support Vector Machines (SVMs) have been widely used in various domains, including medical diagnosis, but they do come with some drawbacks when applied to dental diagnosis:

One significant drawback of SVMs for dental diagnosis is the need for a substantial amount of labeled data. Dental datasets, particularly those with detailed dental images or 3D scans, can be relatively small and expensive to obtain. SVMs perform best when they have a large amount of data to learn from, and this limitation can hinder their effectiveness in dental diagnosis, where data availability is often limited.

Furthermore, SVMs are known to be sensitive to the choice of hyperparameters, such as the kernel function and the regularization parameter. Selecting the appropriate kernel function for dental data can be challenging, as it requires a good understanding of the underlying data distribution. The choice of the regularization parameter can also affect the SVM's performance, and fine-tuning these parameters can be a time-consuming process.

Another drawback of SVMs is their limited ability to handle imbalanced datasets. In dental diagnosis, it's common to encounter imbalanced datasets where certain dental conditions are rare compared to others. SVMs may struggle to correctly classify minority class samples, leading to a biased diagnosis.

Interpretability is another concern with SVMs in dental diagnosis. While they can provide accurate predictions, the models are often considered "black-boxes" in the sense that it can be challenging to interpret why a particular diagnosis was made. In the medical field, especially in dental practice, interpretability is crucial for building trust between healthcare professionals and the model's recommendations.

Lastly, SVMs may not be well-suited for handling high-dimensional dental data efficiently. Dental images, radiographs, or 3D scans can have a large number of features, and SVMs may struggle to handle such high-dimensional data without extensive preprocessing or feature selection, which can introduce additional challenges and potential information loss.

CHAPTER 4

4.PROPOSED METHODOLOGY

4.1 Overview

The procedure for this research work on dental diagnosis involves a series of systematic steps. Firstly, exploratory data analysis (EDA) is conducted to thoroughly investigate the dataset, including statistical summaries and visualizations to identify potential data issues. Following EDA, data normalization is performed to standardize feature scales. The dataset is then divided into training and testing sets to facilitate model training and evaluation. An existing support vector machine (SVM) model is employed as a reference point for comparison with a proposed k-nearest neighbors (KNN) model. Both models are trained and evaluated using the dataset. Finally, the research assesses the performance of these models by employing various metrics to gauge their effectiveness in accurately classifying dental conditions. This structured approach aims to determine the comparative performance of SVM and KNN in the context of dental diagnosis, providing valuable insights into their suitability for this specific application.

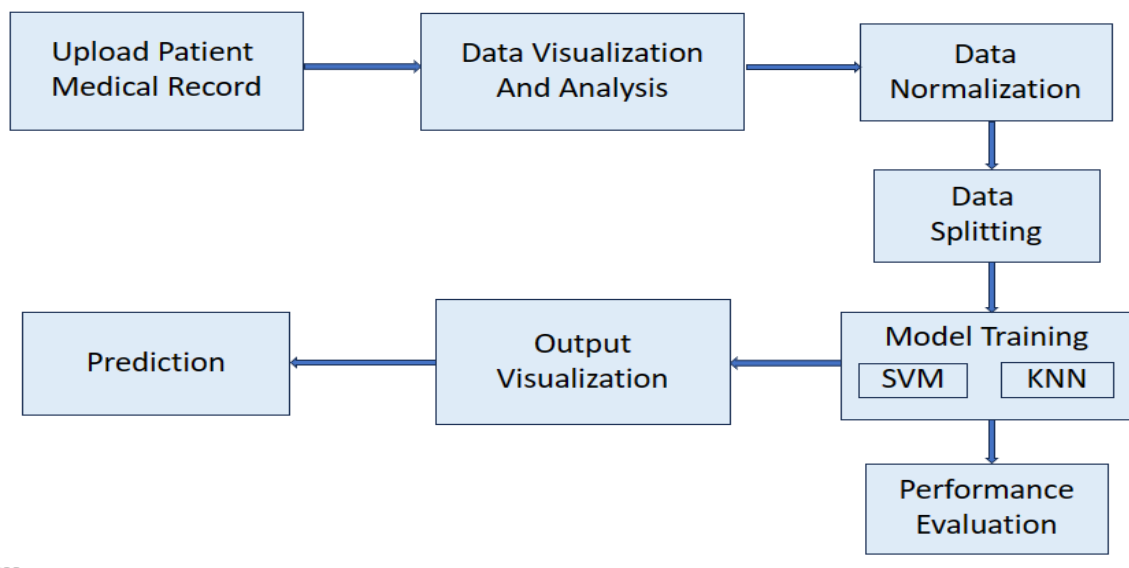


Fig. 4.1: Block diagram of proposed system.

Figure 4.1 shows the proposed system model. The detailed operation illustrated as follows:

EDA (Exploratory Data Analysis): The research work begins with an in-depth exploration of the dental dataset. This step involves summarizing key statistics, visualizing data distributions, and identifying any outliers or missing values. EDA helps researchers gain a comprehensive understanding of the dataset's characteristics and potential challenges.

Data Normalization: After performing EDA, the next step is data normalization. Dental datasets often contain various features with different scales and units. Data normalization is crucial to bring all features to a common scale, typically between 0 and 1, to ensure that the machine learning models can effectively learn from the data without being biased by the magnitude of individual features.

Data Splitting: In preparation for model training and evaluation, the dataset is divided into two or more subsets: a training set and a testing/validation set. The training set is used to train the machine learning models, while the testing/validation set is used to assess their performance. Data splitting helps ensure that the model's performance is evaluated on unseen data, which is a critical step in estimating its real-world applicability.

Existing SVM (Support Vector Machine): In this phase, an existing SVM model is trained on the training dataset. SVM is a popular choice for medical and dental diagnosis tasks due to its ability to handle high-dimensional data and its capacity to find complex decision boundaries. The existing SVM serves as a benchmark or baseline model for comparison with the proposed KNN model.

Proposed KNN (K-Nearest Neighbors): The research introduces a proposed KNN model as an alternative to the existing SVM. KNN is a simple yet effective algorithm for classification tasks. It classifies samples based on the majority class of their k-nearest neighbors. The KNN model is trained and evaluated on the same dataset used for the SVM, and its performance is compared with the SVM's performance.

Performance Prediction: The final step involves predicting and assessing the performance of both the existing SVM and proposed KNN models. Performance metrics such as

accuracy, precision, recall, F1-score, and ROC curves may be used to evaluate how well each model classifies dental conditions. The goal is to determine whether the proposed KNN model outperforms the existing SVM or if there are specific scenarios where one model is more suitable

for dental diagnosis. This step helps provide insights into the efficacy of different machine learning approaches for dental diagnosis tasks.

EDA

The following EDA performed in this work:

Histogram Plots: Histograms are generated for all numeric features in the dataset to visualize the distribution of each variable. This code snippet creates a figure with a size of 15x15, specifies the axis for plotting, and then generates histograms for each numeric feature in the DataFrame 'df.' These histograms help you understand the distribution of values within each feature.

Countplot for Target Variable: A countplot is created to check if the dataset is balanced or not with respect to the 'target' variable. The countplot visualizes the distribution of the 'target' variable, which is typically used in classification tasks to indicate the class labels. By examining the count of each class, you can assess whether the dataset is balanced or skewed towards certain classes.

Correlation Heatmap: A heatmap is generated to visualize the correlation between different features in the dataset. The code calculates the correlation matrix for all features in the DataFrame 'df' and selects the features with the highest correlation. It then plots a heatmap to display the pairwise correlations between these selected features. The 'annot=True' parameter adds numerical values to the heatmap cells, providing insight into the strength and direction of correlations.

4.2 Data Preprocessing

Data pre-processing is a process of preparing the raw data and making it suitable for a machine learning model. It is the first and crucial step while creating a machine learning model. When creating a machine learning project, it is not always a case that we come

across the clean and formatted data. And while doing any operation with data, it is mandatory to clean it and put in a formatted way. So, for this, we use data pre-processing task.

A real-world data generally contains noises, missing values, and maybe in an unusable format which cannot be directly used for machine learning models. Data pre-processing is required tasks for cleaning the data and making it suitable for a machine learning model which also increases the accuracy and efficiency of a machine learning model.

One-Hot Encoding: Categorical variables are one-hot encoded to convert them into a numerical format suitable for machine learning models. The code uses the `pd.get_dummies()` function to create binary columns for each category within categorical variable. This transformation allows machine learning algorithms to work with categorical data effectively.

Standardization: Standard Scaler is applied to scale numeric features, ensuring that they have a mean of 0 and a standard deviation of 1. The 'Standard Scaler' from scikit-learn is used to standardize specific numeric features. Standardization is a common preprocessing step to bring features to a similar scale, which can improve the performance of some machine learning algorithms. This transformation is important for several reasons:

1. **Equal Scaling:** StandardScaler scales each feature to have the same scale. This is crucial for algorithms that are sensitive to the scale of features, such as gradient-based optimization algorithms (e.g., in neural networks) and distance-based algorithms (e.g., k-means clustering).
2. **Mean Centering:** By subtracting the mean from each data point, StandardScaler centers the data around zero. This can help algorithms converge faster during training and improve their performance.
3. **Normalization:** Scaling by the standard deviation normalizes the data, ensuring that features have comparable variances. This can prevent certain features from dominating others in the modeling process.
4. **Interpretability:** Standardized data is more interpretable because it puts all features on a common scale, making it easier to compare the relative importance of features.

4.3 Dataset Splitting

In machine learning data pre-processing, we divide our dataset into a training set and test set. This is one of the crucial steps of data pre-processing as by doing this, we can enhance the performance of our machine learning model. Suppose if we have given training to our machine learning model by a dataset and we test it by a completely different dataset. Then, it will create difficulties for our

model to understand the correlations between the models. If we train our model very well and its training accuracy is also very high, but we provide a new dataset to it, then it will decrease the performance. So we always try to make a machine learning model which performs well with the training set and also with the test dataset.

Training Set: A subset of dataset to train the machine learning model, and we already know the output.

Test set: A subset of dataset to test the machine learning model, and by using the test set, model predicts the output.

4.4 KNN

K-Nearest Neighbors (KNN) is a simple yet powerful supervised machine learning algorithm used for classification and regression tasks. It's based on the idea that data points with similar features tend to belong to the same class or have similar values in the case of regression. KNN is a distance-based classification algorithm. It assigns a new data point to the majority class of its k-nearest neighbors. The choice of 'k' (the number of neighbors) is a crucial hyperparameter that impacts the model's performance. KNN is an instance-based learning method, meaning it doesn't build a model during training. Instead, it memorizes the entire training dataset and uses it for predictions.

Working Principle:

Step 1: Distance Metric:

- KNN uses a distance metric (typically Euclidean distance, but others like Manhattan, Minkowski, etc., are also possible) to measure the similarity between data points. The algorithm finds the 'k' nearest neighbors with the smallest distances to the new data point.

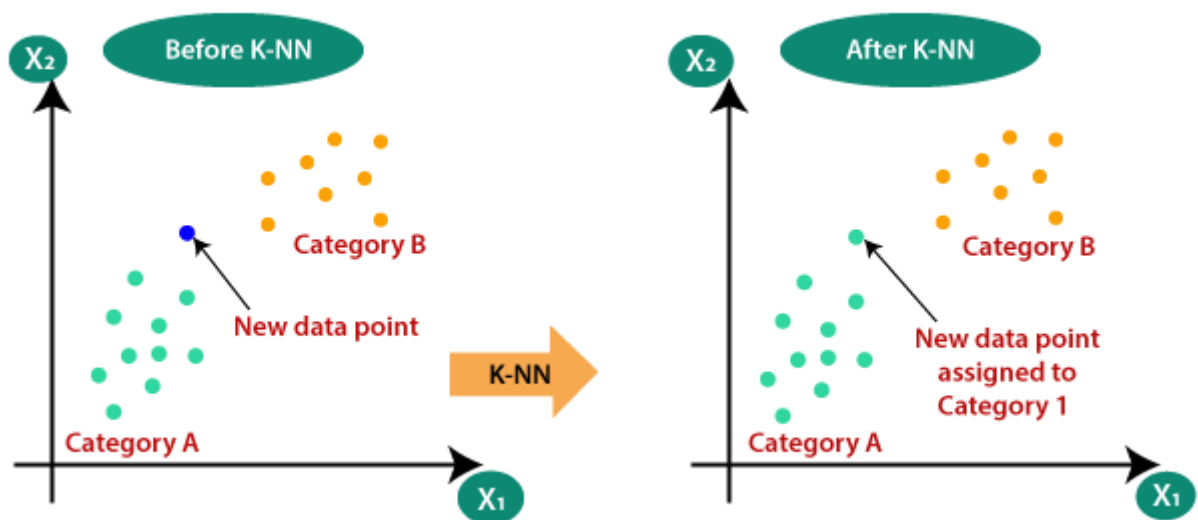


Figure 4.2. KNN initialization

- Voting Mechanism: For classification, KNN uses a majority voting mechanism among its neighbors. The class that occurs most frequently among the neighbors is assigned to the new data point. For regression, it takes the mean (or median) value of the 'k' nearest neighbors as the prediction.

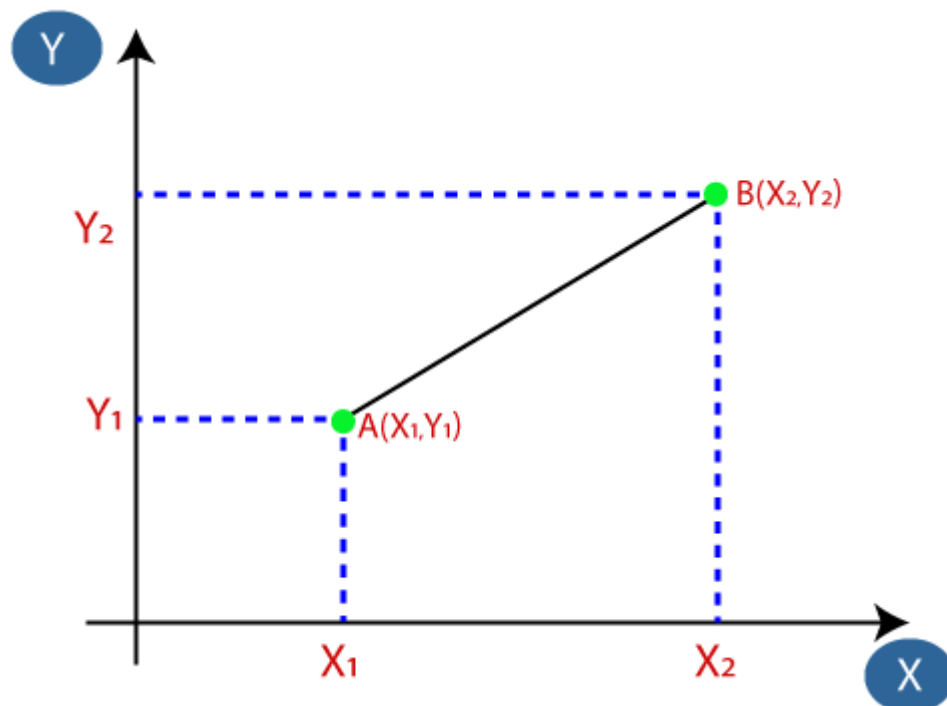


Figure 4.2. Distance measurement in KNN.

Step 2. Hyperparameter 'k':

- **Choosing the Right 'k':** The choice of 'k' is crucial. A small 'k' makes the model sensitive to noise and outliers but may capture local patterns well. A large 'k' smooths out local variations but can make the model less accurate.
- **Methods for Choosing 'k':** Cross-validation, grid search, and domain knowledge are common approaches to determine the optimal 'k' value.
- **Simplicity:** KNN is easy to understand and implement, making it a suitable choice for beginners.
- **No Training Phase:** It doesn't require a training phase since it memorizes the data, making it suitable for online learning and non-stationary data.
- **Non-Parametric:** KNN is non-parametric, meaning it makes no assumptions about the underlying data distribution.
- **Works for Multiclass Problems:** KNN naturally handles multi-class classification problems.

Step 3. Variants:

- **Weighted KNN:** Assigns different weights to neighbors based on their distance. Closer neighbors have a greater influence on the prediction.
- **KNN with Feature Scaling:** Feature scaling is essential when using KNN, as it's distance-based. Standardization (scaling features to have mean=0 and standard deviation=1) is often applied.
- **KD-Tree and Ball-Tree:** These data structures can be used to speed up KNN search for large datasets.

4.5 Advantages of KNN

- **Simplicity:** KNN is a simple and easy-to-understand algorithm. It does not make strong assumptions about the underlying data distribution, making it versatile and applicable to various types of datasets.

-
-
- **No Training Time:** Unlike many other machine learning algorithms that require training on a large dataset, KNN does not have a training phase. It stores the entire dataset and makes predictions based on the proximity of new data points to existing ones.
 - **Versatility:** KNN can be used for both classification and regression tasks. It can handle categorical and numerical data, making it suitable for a wide range of problems.
 - **Interpretability:** KNN's decision-making process is intuitive. Predictions are made based on the majority class of the k-nearest neighbors, which can be easily explained and understood.
 - **Adaptability:** KNN can adapt to changes in the dataset since it doesn't assume any specific functional form. This makes it suitable for scenarios where the data distribution is not stationary.
-
- **Robustness to Outliers:** KNN is robust to outliers in the data. Outliers do not significantly impact its performance since predictions are based on the majority class of the neighbors.
 - **Non-Parametric:** KNN is considered a non-parametric algorithm because it doesn't make explicit assumptions about the data's underlying distribution. It doesn't impose a fixed structure on the data.
 - **Local Learning:** KNN focuses on local patterns within the data, making it effective when the decision boundary is not globally linear or when data clusters in certain regions of the feature space.
 - **Ease of Implementation:** Implementing KNN is straightforward, and it doesn't require complex optimization procedures. This simplicity makes it a good choice for quick prototyping.
 - **Ensemble Potential:** KNN can be part of an ensemble learning approach, such as bagging or boosting, to further improve its predictive performance.

CHAPTER 5

UML DAIGRAMS

UML stands for Unified Modelling Language. UML is a standardized general-purpose modelling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group. The goal is for UML to become a common language for creating models of object-oriented computer software. In its current form UML is comprised of two major components: a Meta-model and a notation. In the future, some form of method or process may also be added to; or associated with, UML.

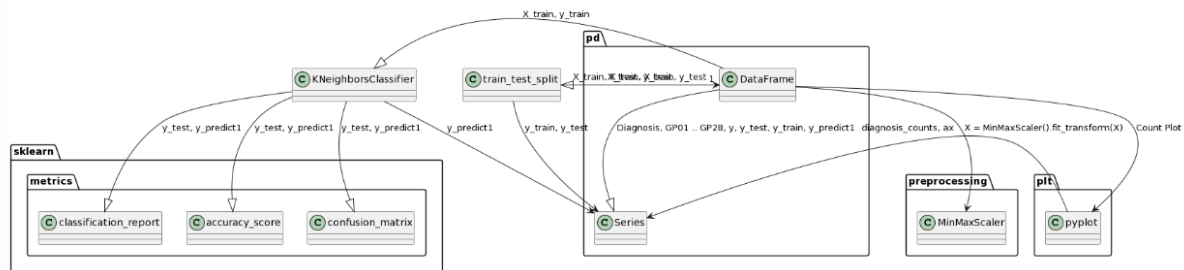
The Unified Modelling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modelling and other non-software systems. The UML represents a collection of best engineering practices that have proven successful in the modelling of large and complex systems. The UML is a very important part of developing objects-oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

GOALS: The Primary goals in the design of the UML are as follows:

- Provide users a ready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models.
- Provide extendibility and specialization mechanisms to extend the core concepts.
- Be independent of particular programming languages and development process.
- Provide a formal basis for understanding the modeling language.
- Encourage the growth of OO tools market.
- Support higher level development concepts such as collaborations, frameworks, patterns and components.
- Integrate best practices.

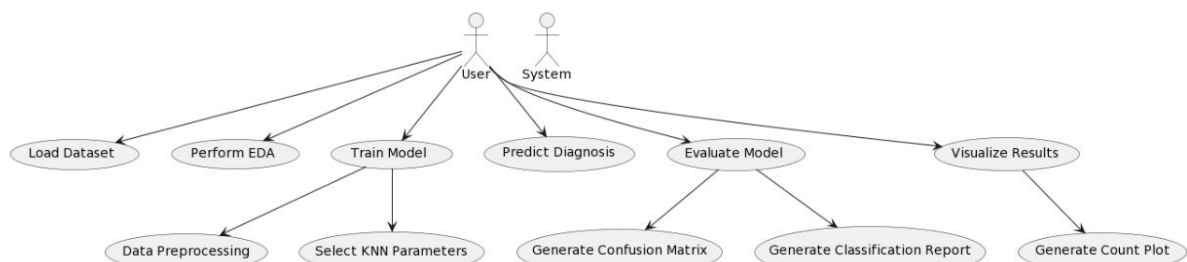
Class diagram

The class diagram is used to refine the use case diagram and define a detailed design of the system. The class diagram classifies the actors defined in the use case diagram into a set of interrelated classes. The relationship or association between the classes can be either an "is-a" or "has-a" relationship. Each class in the class diagram may be capable of providing certain functionalities. These functionalities provided by the class are termed "methods" of the class. Apart from this, each class may have certain "attributes" that uniquely identify the class.



Use case Diagram

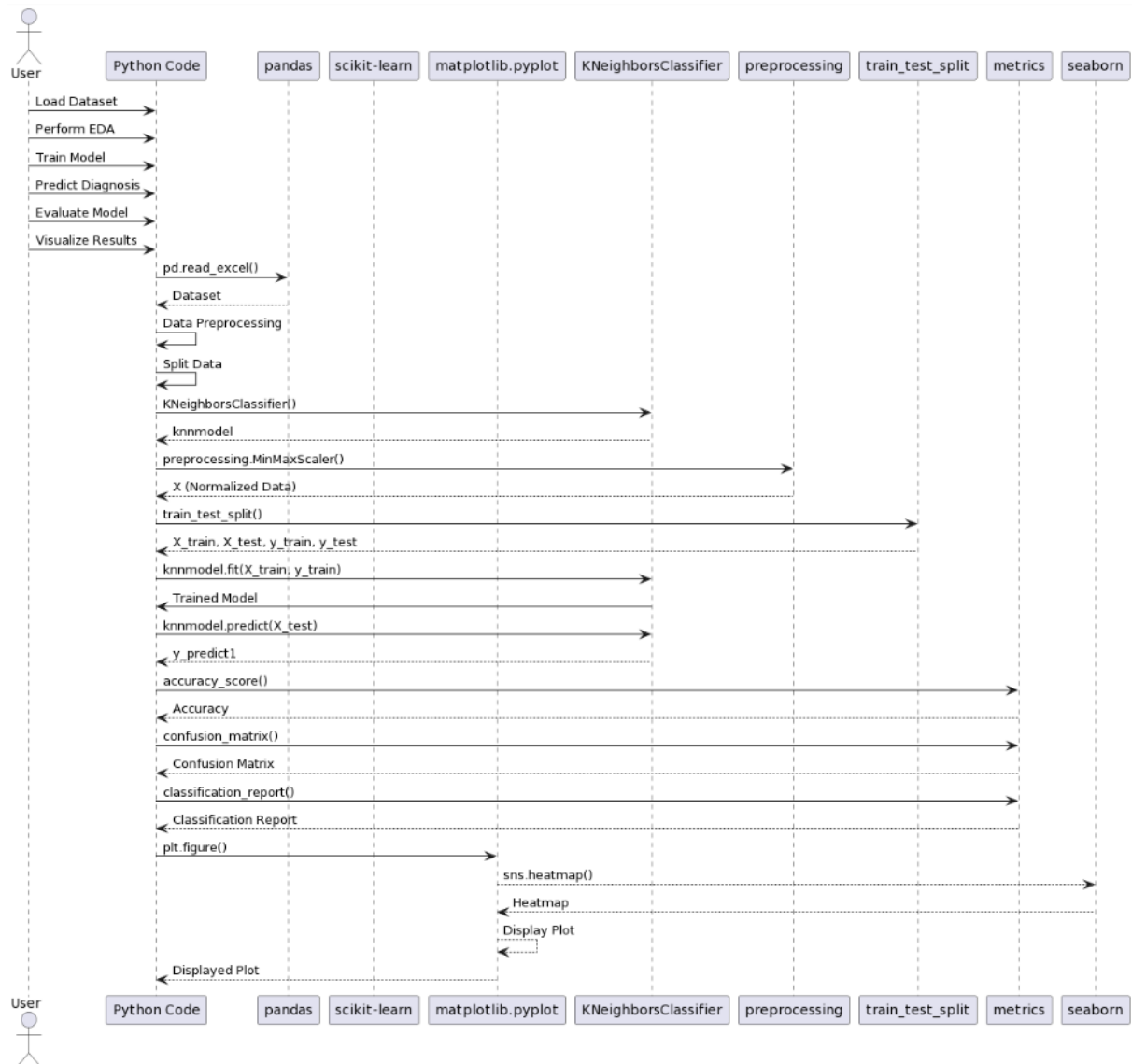
A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.



Sequence Diagram

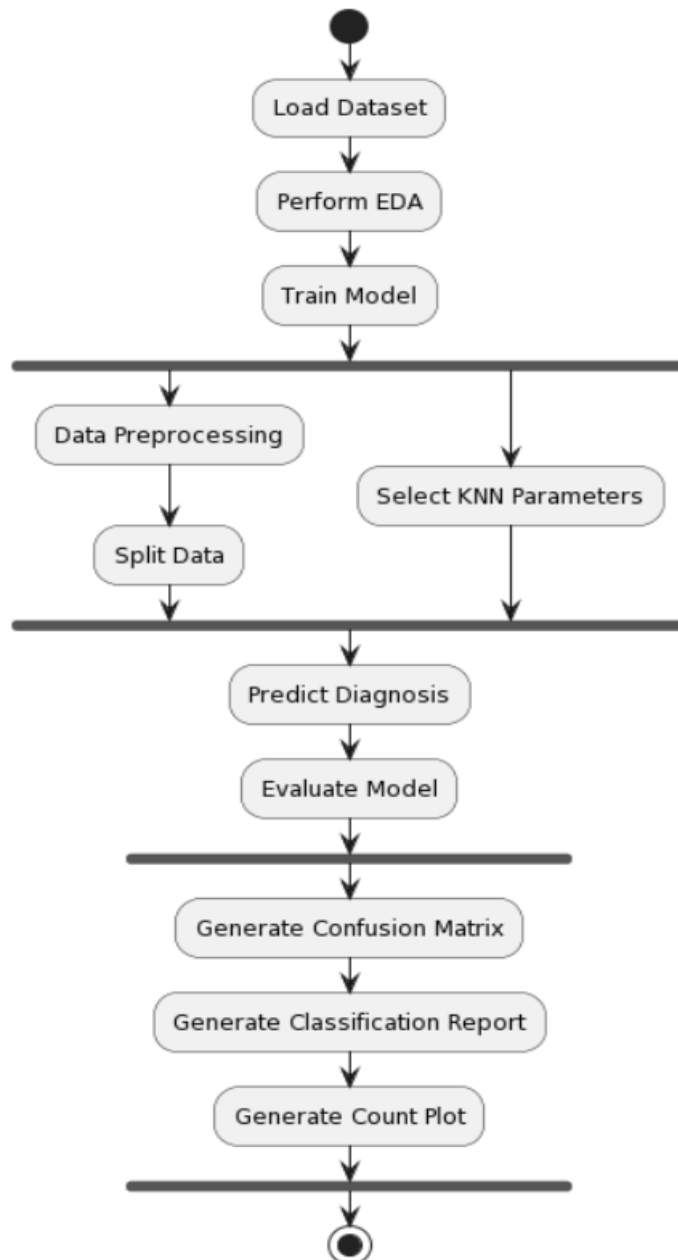
A **sequence diagram** in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. A sequence diagram shows, as parallel vertical lines ("lifelines"), different

processes or objects that live simultaneously, and as horizontal arrows, the messages exchanged between them, in the order in which they occur. This allows the specification of simple runtime scenarios in a graphical manner.



Activity diagram: Activity diagrams are graphical representations of Workflows of stepwise activities and actions with support for choice, iteration, and concurrency.

In the Unified Modeling Language, activity diagrams can be used to describe the business and operational step-by-step workflows of components in a system. An activity diagram shows the overall flow of control.



CHAPTER 6

SOFTWARE ENVIRONMENT

What is Python?

Below are some facts about Python.

- Python is currently the most widely used multi-purpose, high-level programming language.
- Python allows programming in Object-Oriented and Procedural paradigms. Python programs generally are smaller than other programming languages like Java.
- Programmers have to type relatively less and indentation requirement of the language, makes them readable all the time.
- Python language is being used by almost all tech-giant companies like – Google, Amazon, Facebook, Instagram, Dropbox, Uber... etc.

The biggest strength of Python is huge collection of standard libraries which can be used for the following –

- Machine Learning
- GUI Applications (like Kivy, Tkinter, PyQt etc.)
- Web frameworks like Django (used by YouTube, Instagram, Dropbox)
- Image processing (like Opencv, Pillow)
- Web scraping (like Scrapy, BeautifulSoup, Selenium)
- Test frameworks
- Multimedia

Advantages of Python

Let's see how Python dominates over other languages.

1. Extensive Libraries

Python downloads with an extensive library and it contain code for various purposes like regular expressions, documentation-generation, unit-testing, web browsers, threading, databases, CGI, email, image manipulation, and more. So, we don't have to write the complete code for that manually.

2. Extensible

As we have seen earlier, Python can be extended to other languages. You can write some of your code in languages like C++ or C. This comes in handy, especially in projects.

3. Embeddable

Complimentary to extensibility, Python is embeddable as well. You can put your Python code in your source code of a different language, like C++. This lets us add scripting capabilities to our code in the other language.

4. Improved Productivity

The language's simplicity and extensive libraries render programmers more productive than languages like Java and C++ do. Also, the fact that you need to write less and get more things done.

5. IOT Opportunities

Since Python forms the basis of new platforms like Raspberry Pi, it finds the future bright for the Internet of Things. This is a way to connect the language with the real world.

6. Simple and Easy

When working with Java, you may have to create a class to print 'Hello World'. But in Python, just a print statement will do. It is also quite easy to learn, understand, and code. This is why when people pick up Python, they have a hard time adjusting to other more verbose languages like Java.

7. Readable

Because it is not such a verbose language, reading Python is much like reading English. This is the reason why it is so easy to learn, understand, and code. It also does not need curly braces to define blocks, and indentation is mandatory. These further aids the readability of the code.

8. Object-Oriented

This language supports both the procedural and object-oriented programming paradigms. While functions help us with code reusability, classes and objects let us model the real world. A class allows the encapsulation of data and functions into one.

9. Free and Open-Source

Like we said earlier, Python is freely available. But not only can you download Python for free, but you can also download its source code, make changes to it, and even distribute it. It downloads with an extensive collection of libraries to help you with your tasks.

10. Portable

When you code your project in a language like C++, you may need to make some changes to it if you want to run it on another platform. But it isn't the same with Python. Here, you need to code only once, and you can run it anywhere. This is called Write Once Run Anywhere (WORA). However, you need to be careful enough not to include any system-dependent features.

11. Interpreted

Lastly, we will say that it is an interpreted language. Since statements are executed one by one, debugging is easier than in compiled languages.

Any doubts till now in the advantages of Python? Mention in the comment section.

Advantages of Python Over Other Languages

1. Less Coding

Almost all of the tasks done in Python requires less coding when the same task is done in other languages. Python also has an awesome standard library support, so you don't have

to search for any third-party libraries to get your job done. This is the reason that many people suggest learning Python to beginners.

2. Affordable

Python is free therefore individuals, small companies or big organizations can leverage the free available resources to build applications. Python is popular and widely used so it gives you better community support.

The 2019 Github annual survey showed us that Python has overtaken Java in the most popular programming language category.

3. Python is for Everyone

Python code can run on any machine whether it is Linux, Mac or Windows. Programmers need to learn different languages for different jobs but with Python, you can professionally build web apps, perform data analysis and machine learning, automate things, do web scraping and also build games and powerful visualizations. It is an all-rounder programming language.

Disadvantages of Python

So far, we've seen why Python is a great choice for your project. But if you choose it, you should be aware of its consequences as well. Let's now see the downsides of choosing Python over another language.

1. Speed Limitations

We have seen that Python code is executed line by line. But since Python is interpreted, it often results in slow execution. This, however, isn't a problem unless speed is a focal point for the project. In other words, unless high speed is a requirement, the benefits offered by Python are enough to distract us from its speed limitations.

2. Weak in Mobile Computing and Browsers

While it serves as an excellent server-side language, Python is much rarely seen on the client-side. Besides that, it is rarely ever used to implement smartphone-based applications. One such application is called Carbonnelle.

The reason it is not so famous despite the existence of Brython is that it isn't that secure.

3. Design Restrictions

As you know, Python is dynamically-typed. This means that you don't need to declare the type of variable while writing the code. It uses duck-typing. But wait, what's that? Well, it just means that

if it looks like a duck, it must be a duck. While this is easy on the programmers during coding, it can raise run-time errors.

4. Underdeveloped Database Access Layers

Compared to more widely used technologies like JDBC (Java DataBase Connectivity) and ODBC (Open DataBase Connectivity), Python's database access layers are a bit underdeveloped. Consequently, it is less often applied in huge enterprises.

5. Simple

No, we're not kidding. Python's simplicity can indeed be a problem. Take my example. I don't do Java, I'm more of a Python person. To me, its syntax is so simple that the verbosity of Java code seems unnecessary.

This was all about the Advantages and Disadvantages of Python Programming Language.

History of Python

What do the alphabet and the programming language Python have in common? Right, both start with ABC. If we are talking about ABC in the Python context, it's clear that the programming language ABC is meant. ABC is a general-purpose programming language and programming environment, which had been developed in the Netherlands, Amsterdam, at the CWI (Centrum Wiskunde & Informatica). The greatest achievement of ABC was to influence the design of Python. Python was conceptualized in the late 1980s. Guido van Rossum worked that time in a project at the CWI, called Amoeba, a distributed operating

system. In an interview with Bill Venner¹, Guido van Rossum said: "In the early 1980s, I worked as an implementer on a team building a language called ABC at Centrum voor Wiskunde en Informatica (CWI). I don't know how well people know ABC's influence on Python. I try to mention ABC's influence because I'm indebted to everything I learned during that project and to the people who worked on it." Later on in the same Interview, Guido van Rossum continued: "I remembered all my experience and some of my frustration with ABC. I decided to try to design a simple scripting language that possessed some of ABC's better properties, but without its problems. So, I started typing. I created a simple virtual machine, a simple parser, and a simple runtime. I made my own version of the various ABC parts that I liked. I created a basic syntax,

used indentation for statement grouping instead of curly braces or begin-end blocks, and developed a small number of powerful data types: a hash table (or dictionary, as we call it), a list, strings, and numbers."

Python Development Steps

Guido Van Rossum published the first version of Python code (version 0.9.0) at alt.sources in February 1991. This release included already exception handling, functions, and the core data types of lists, dict, str and others. It was also object oriented and had a module system. Python version 1.0 was released in January 1994. The major new features included in this release were the functional programming tools lambda, map, filter and reduce, which Guido Van Rossum never liked. Six and a half years later in October 2000, Python 2.0 was introduced. This release included list comprehensions, a full garbage collector and it was supporting unicode. Python flourished for another 8 years in the versions 2.x before the next major release as Python 3.0 (also known as "Python 3000" and "Py3K") was released. Python 3 is not backwards compatible with Python 2.x. The emphasis in Python 3 had been on the removal of duplicate programming constructs and modules, thus fulfilling or coming close to fulfilling the 13th law of the Zen of Python: "There should be one -- and preferably only one -- obvious way to do it." Some changes in Python 7.3:

- Print is now a function.
- Views and iterators instead of lists
- The rules for ordering comparisons have been simplified. E.g., a heterogeneous list

cannot be sorted, because all the elements of a list must be comparable to each other.

- There is only one integer type left, i.e., int. long is int as well.
- The division of two integers returns a float instead of an integer. "/" can be used to have the "old" behaviour.
- Text Vs. Data Instead of Unicode Vs. 8-bit

Purpose

We demonstrated that our approach enables successful segmentation of intra-retinal layers—even with low-quality images containing speckle noise, low contrast, and different intensity ranges throughout—with the assistance of the ANIS feature.

Python

Python is an interpreted high-level programming language for general-purpose programming. Created by Guido van Rossum and first released in 1991, Python has a design philosophy that emphasizes code readability, notably using significant whitespace.

Python features a dynamic type system and automatic memory management. It supports multiple programming paradigms, including object-oriented, imperative, functional and procedural, and has a large and comprehensive standard library.

- Python is Interpreted – Python is processed at runtime by the interpreter. You do not need to compile your program before executing it. This is similar to PERL and PHP.
- Python is Interactive – you can actually sit at a Python prompt and interact with the interpreter directly to write your programs.

Python also acknowledges that speed of development is important. Readable and terse code is part of this, and so is access to powerful constructs that avoid tedious repetition of code. Maintainability also ties into this may be an all but useless metric, but it does say something about how much code you have to scan, read and/or understand to troubleshoot problems or tweak behaviors. This speed of development, the ease with which a programmer of other languages can pick up basic Python skills and the huge standard library is key to another

area where Python excels. All its tools have been quick to implement, saved a lot of time, and several of them have later been patched and updated by people with no Python background - without breaking.

Modules Used in Project

TensorFlow

TensorFlow is a free and open-source software library for dataflow and differentiable programming across a range of tasks. It is a symbolic math library and is also used for machine learning applications such as neural networks. It is used for both research and production at Google.

TensorFlow was developed by the Google Brain team for internal Google use. It was released under the Apache 2.0 open-source license on November 9, 2015.

NumPy

NumPy is a general-purpose array-processing package. It provides a high-performance multidimensional array object, and tools for working with these arrays.

It is the fundamental package for scientific computing with Python. It contains various features including these important ones:

- A powerful N-dimensional array object
- Sophisticated (broadcasting) functions
- Tools for integrating C/C++ and Fortran code
- Useful linear algebra, Fourier transform, and random number capabilities

Besides its obvious scientific uses, NumPy can also be used as an efficient multi-dimensional container of generic data. Arbitrary datatypes can be defined using NumPy which allows NumPy to seamlessly and speedily integrate with a wide variety of databases.

Pandas

Pandas is an open-source Python Library providing high-performance data manipulation and analysis tool using its powerful data structures. Python was majorly used for data munging and preparation. It had very little contribution towards data analysis. Pandas solved this problem. Using Pandas, we can accomplish five typical steps in the processing and analysis of data, regardless of the origin of data load, prepare, manipulate, model, and analyze. Python with Pandas is used in a wide range of fields including academic and commercial domains including finance, economics, Statistics, analytics, etc.

Matplotlib

Matplotlib is a Python 2D plotting library which produces publication quality figures in a variety of hardcopy formats and interactive environments across platforms. Matplotlib can be used in

Python scripts, the Python and IPython shells, the Jupyter Notebook, web application servers, and four graphical user interface toolkits. Matplotlib tries to make easy things easy and hard things possible. You can generate plots, histograms, power spectra, bar charts, error charts, scatter plots, etc., with just a few lines of code. For examples, see the sample plots and thumbnail gallery.

For simple plotting the pyplot module provides a MATLAB-like interface, particularly when combined with IPython. For the power user, you have full control of line styles, font properties, axes properties, etc, via an object-oriented interface or via a set of functions familiar to MATLAB users.

Scikit – learn

Scikit-learn provides a range of supervised and unsupervised learning algorithms via a consistent interface in Python. It is licensed under a permissive simplified BSD license and is distributed under many Linux distributions, encouraging academic and commercial use. Python

Python is an interpreted high-level programming language for general-purpose programming. Created by Guido van Rossum and first released in 1991, Python has a design philosophy that emphasizes code readability, notably using significant whitespace.

Python features a dynamic type system and automatic memory management. It supports multiple programming paradigms, including object-oriented, imperative, functional and procedural, and has a large and comprehensive standard library.

- Python is Interpreted – Python is processed at runtime by the interpreter. You do not need to compile your program before executing it. This is similar to PERL and PHP.
- Python is Interactive – you can actually sit at a Python prompt and interact with the interpreter directly to write your programs.

Python also acknowledges that speed of development is important. Readable and terse code is part of this, and so is access to powerful constructs that avoid tedious repetition of code. Maintainability also ties into this may be an all but useless metric, but it does say something about how much code you have to scan, read and/or understand to troubleshoot problems or tweak behaviors. This speed

of development, the ease with which a programmer of other languages can pick up basic Python skills and the huge standard library is key to another area where Python excels. All its tools have been quick to implement, saved a lot of time, and several of them have later been patched and updated by people with no Python background - without breaking.

Install Python Step-by-Step in Windows and Mac

Python a versatile programming language doesn't come pre-installed on your computer devices. Python was first released in the year 1991 and until today it is a very popular high-level programming language. Its style philosophy emphasizes code readability with its notable use of great whitespace.

The object-oriented approach and language construct provided by Python enables programmers to write both clear and logical code for projects. This software does not come pre-packaged with Windows.

How to Install Python on Windows and Mac

There have been several updates in the Python version over the years. The question is how to install Python? It might be confusing for the beginner who is willing to start learning Python but this tutorial will solve your query. The latest or the newest version of Python is version 3.7.4 or in other words, it is Python 3.

Note: The python version 3.7.4 cannot be used on Windows XP or earlier devices.

Before you start with the installation process of Python. First, you need to know about your System Requirements. Based on your system type i.e., operating system and based processor, you must download the python version. My system type is a Windows 64-bit operating system. So, the steps below are to install python version 3.7.4 on Windows 7 device or to install Python 3. Download the Python Cheatsheet here. The steps on how to install Python on Windows 10, 8 and 7 are divided into 4 parts to help understand better.

Download the Correct version into the system

Step 1: Go to the official site to download and install python using Google Chrome or any other web browser. OR Click on the following link: <https://www.python.org>



Now, check for the latest and the correct version for your operating system.

Step 2: Click on the Download Tab.



Step 3: You can either select the Download Python for windows 3.7.4 button in Yellow Color or you can scroll further down and click on download with respective to their version. Here, we are downloading the most recent python version for windows 3.7.4

Looking for a specific release?			
Python releases by version number:			
Release version	Release date		Click for more
Python 3.7.4	July 8, 2019	 Download	Release Notes
Python 3.6.9	July 2, 2019	 Download	Release Notes
Python 3.7.3	March 25, 2019	 Download	Release Notes
Python 3.4.10	March 18, 2019	 Download	Release Notes
Python 3.5.7	March 18, 2019	 Download	Release Notes
Python 3.7.16	March 4, 2019	 Download	Release Notes
Python 3.7.2	Dec. 24, 2018	 Download	Release Notes

Step 4: Scroll down the page until you find the Files option.

Step 5: Here you see a different version of python along with the operating system.

Files					
Version	Operating System	Description	MD5 Sum	File Size	GPG
Gzipped source tarball	Source release		68111671e5b2db4ae77b9ab01b0f9be	23017663	50G
XZ compressed source tarball	Source release		d33e4aa66097051c2eca45ee3604803	17131432	50G
macOS 64-bit/32-bit installer	Mac OS X	for Mac OS X 10.6 and later	6428b4fa7583da7f1a4c2ba1ce08e6	34898416	50G
macOS 64-bit installer	Mac OS X	for OS X 10.9 and later	5dd605c38217a45773bf5e4a936b241f	28082845	50G
Windows .hug file	Windows		d63999573a2c96b2ac56cade6b4f7cd2	8131761	50G
Windows x86-64 embeddable zip file	Windows	for AMD64/EM64T/x64	9600c3c7d29e0b9abe63184a40729a2	7504391	50G
Windows x86-64 executable installer	Windows	for AMD64/EM64T/x64	a702b4b0a7f0d4b0b3543a583e543400	26880368	50G
Windows x86-64 web-based installer	Windows	for AMD64/EM64T/x64	28c61c608b6d73a69e53a3bd351b4bd2	1362904	50G
Windows x86 embeddable zip file	Windows		9fab3bd18a41879fa94133574139d8	6741626	50G
Windows x86 executable installer	Windows		33cc882942a54446a3d645147e194789	25663848	50G
Windows x86 web-based installer	Windows		1b670cfa5d317d82c30983ea371d87c	1324608	50G

- To download Windows 32-bit python, you can select any one from the three options: Windows x86 embeddable zip file, Windows x86 executable installer or Windows x86 web-based installer.
- To download Windows 64-bit python, you can select any one from the three options: Windows x86-64 embeddable zip file, Windows x86-64 executable installer or Windows

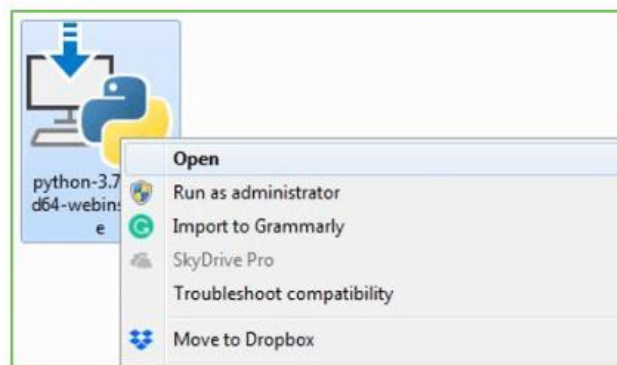
x86-64 web-based installer.

Here we will install Windows x86-64 web-based installer. Here your first part regarding which version of python is to be downloaded is completed. Now we move ahead with the second part in installing python i.e., Installation

Note: To know the changes or updates that are made in the version you can click on the Release Note Option.

Installation of Python

Step 1: Go to Download and Open the downloaded python version to carry out the installation process.



Step 2: Before you click on Install Now, make sure to put a tick on Add Python 3.7 to PATH.



Step 3: Click on Install NOW After the installation is successful. Click on Close.



With these above three steps on python installation, you have successfully and correctly installed Python. Now is the time to verify the installation.

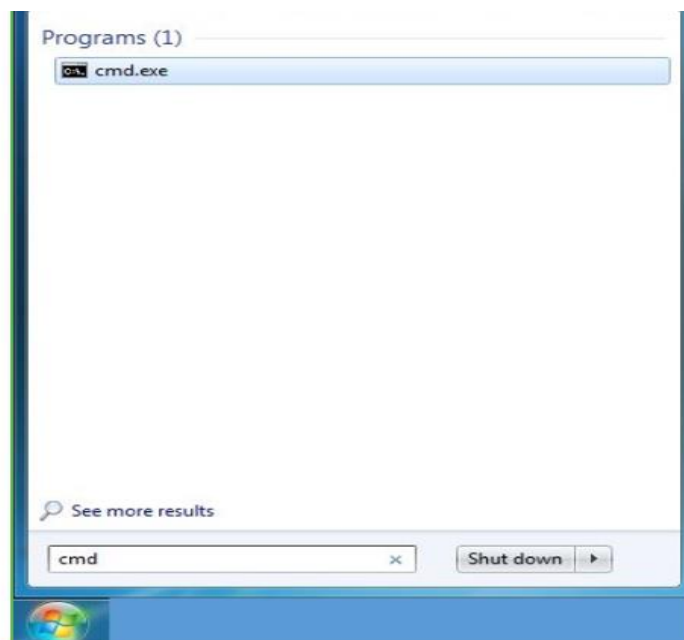
Note: The installation process might take a couple of minutes.

Verify the Python Installation

Step 1: Click on Start

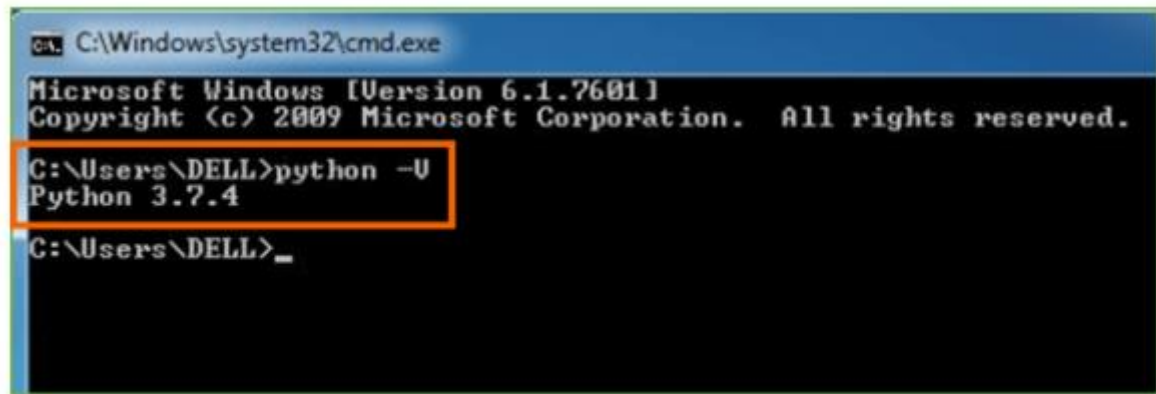
Step

2: In the Windows Run Command, type “cmd”.



Step 3: Open the Command prompt option.

Step 4: Let us test whether the python is correctly installed. Type python -V and press Enter.



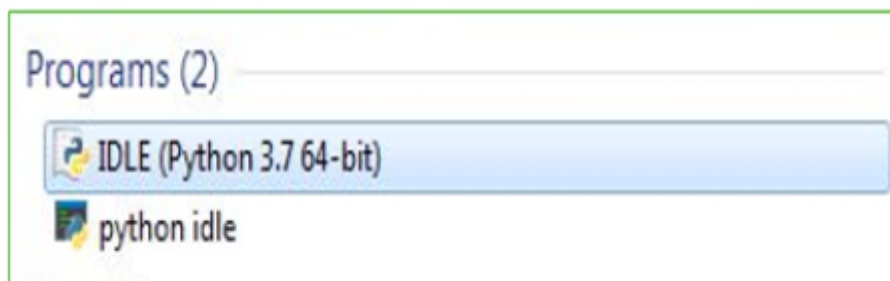
Step 5: You will get the answer as 3.7.4

Note: If you have any of the earlier versions of Python already installed. You must first uninstall the earlier version and then install the new one.

Check how the Python IDLE works

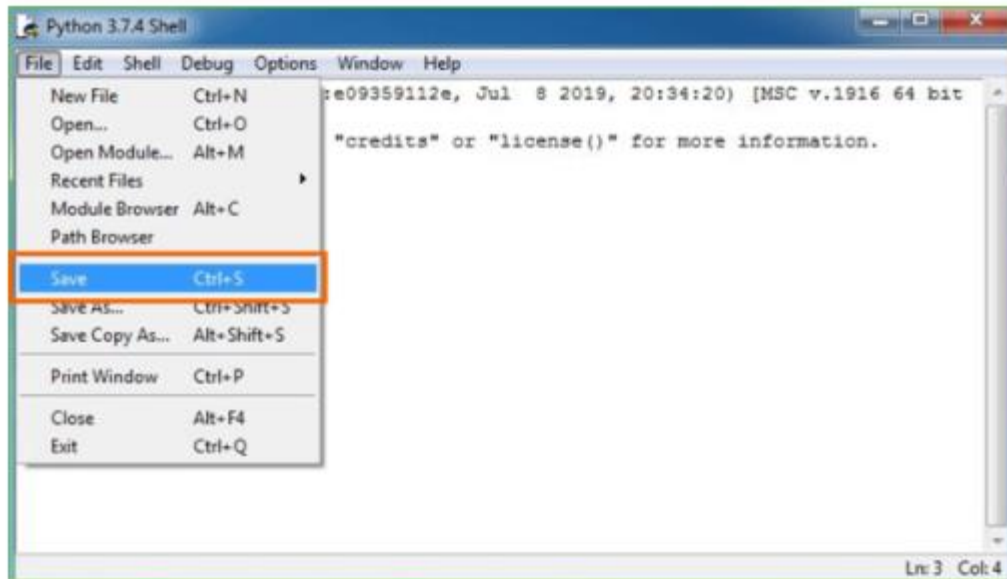
Step 1: Click on Start

Step 2: In the Windows Run command, type “python idle”.



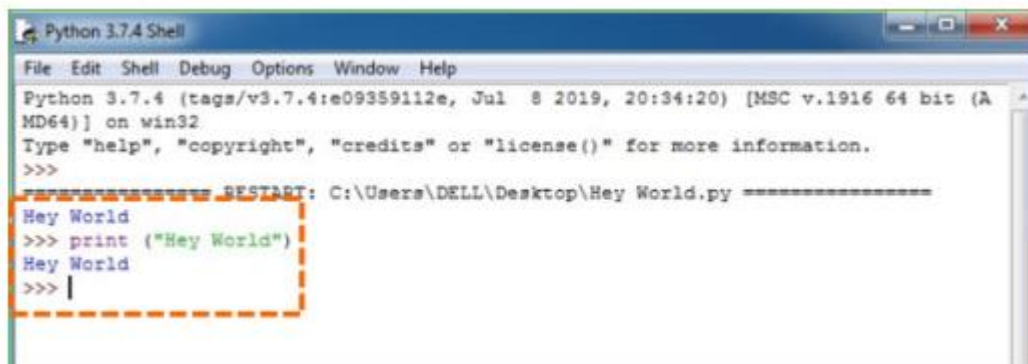
Step 3: Click on IDLE (Python 3.7 64-bit) and launch the program

Step 4: To go ahead with working in IDLE you must first save the file. Click on File > Click on Save



Step 5: Name the file and save as type should be Python files. Click on SAVE. Here I have named the files as Hey World.

Step 6: Now for e.g., enter print ("Hey World") and Press Enter.



You will see that the command given is launched. With this, we end our tutorial on how to install Python. You have learned how to download python for windows into your respective operating system.

Note: Unlike Java, Python does not need semicolons at the end of the statements otherwise it won't work.

CHAPTER 7

SYSTEM REQUIREMENTS SPECIFICATIONS

Software Requirements

The functional requirements or the overall description documents include the product perspective and features, operating system and operating environment, graphics requirements, design constraints and user documentation.

The appropriation of requirements and implementation constraints gives the general overview of the project in regard to what the areas of strength and deficit are and how to tackle them.

- Python IDLE 3.7 version (or)
- Anaconda 3.7 (or)
- Jupiter (or)
- Google colab

Hardware Requirements

Minimum hardware requirements are very dependent on the particular software being developed by a given Enthought Python / Canopy / VS Code user. Applications that need to store large arrays/objects in memory will require more RAM, whereas applications that need to perform numerous calculations or tasks more quickly will require a faster processor.

Operating system	:	Windows, Linux
Processor	:	minimum intel i3
Ram	:	minimum 4 GB
Hard disk	:	minimum 250GB

CHAPTER 8

FUNCTIONAL REQUIREMENTS

Output Design

Outputs from computer systems are required primarily to communicate the results of processing to users. They are also used to provide a permanent copy of the results for later consultation. The various types of outputs in general are:

- External Outputs, whose destination is outside the organization
- Internal Outputs whose destination is within organization and they are the
- User's main interface with the computer.
- Operational outputs whose use is purely within the computer department.
- Interface outputs, which involve the user in communicating directly.

Output Definition

The outputs should be defined in terms of the following points:

- Type of the output
- Content of the output
- Format of the output
- Location of the output
- Frequency of the output
- Volume of the output
- Sequence of the output

It is not always desirable to print or display data as it is held on a computer. It should be decided as which form of the output is the most suitable.

Input Design

Input design is a part of overall system design. The main objective during the input design is as given below:

- To produce a cost-effective method of input.
- To achieve the highest possible level of accuracy.

-
- To ensure that the input is acceptable and understood by the user.

Input Stages

The main input stages can be listed as below:

- Data recording
- Data transcription
- Data conversion
- Data verification
- Data control
- Data transmission
- Data validation
- Data correction

Input Types

It is necessary to determine the various types of inputs. Inputs can be categorized as follows:

- External inputs, which are prime inputs for the system.
- Internal inputs, which are user communications with the system.
- Operational, which are computer department's communications to the system?
- Interactive, which are inputs entered during a dialogue.

Input Media

At this stage choice has to be made about the input media. To conclude about the input media consideration has to be given to;

- Type of input
- Flexibility of format
- Speed
- Accuracy
- Verification methods

- Rejection rates
- Ease of correction
- Storage and handling requirements

-
- Security
 - Easy to use
 - Portability

Keeping in view the above description of the input types and input media, it can be said that most of the inputs are of the form of internal and interactive. As

Input data is to be the directly keyed in by the user, the keyboard can be considered to be the most suitable input device.

Error Avoidance

At this stage care is to be taken to ensure that input data remains accurate from the stage at which it is recorded up to the stage in which the data is accepted by the system. This can be achieved only by means of careful control each time the data is handled.

Error Detection

Even though every effort is made to avoid the occurrence of errors, still a small proportion of errors is always likely to occur, these types of errors can be discovered by using validations to check the input data.

Data Validation

Procedures are designed to detect errors in data at a lower level of detail. Data validations have been included in the system in almost every area where there is a possibility for the user to commit errors. The system will not accept invalid data. Whenever an invalid data is keyed in, the system immediately prompts the user and the user has to again key in the data and the system will accept the data only if the data is correct. Validations have been included where necessary.

The system is designed to be a user friendly one. In other words the system has been designed to communicate effectively with the user. The system has been designed with popup menus.

User Interface Design

It is essential to consult the system users and discuss their needs while designing the user interface:

User Interface Systems Can Be Broadly Classified As:

- User initiated interface the user is in charge, controlling the progress of the user/computer dialogue. In the computer-initiated interface, the computer selects the next stage in the interaction.
- Computer initiated interfaces

In the computer-initiated interfaces the computer guides the progress of the user/computer dialogue. Information is displayed and the user response of the computer takes action or displays further information.

User Initiated Interfaces

User initiated interfaces fall into two approximate classes:

- Command driven interfaces: In this type of interface the user inputs commands or queries which are interpreted by the computer.
- Forms oriented interface: The user calls up an image of the form to his/her screen and fills in the form. The forms-oriented interface is chosen because it is the best choice.

Computer-Initiated Interfaces

The following computer – initiated interfaces were used:

- The menu system for the user is presented with a list of alternatives and the user chooses one; of alternatives.
- Questions – answer type dialog system where the computer asks question and takes action based on the basis of the users reply.

Right from the start the system is going to be menu driven, the opening menu displays the available options. Choosing one option gives another popup menu with more options. In this way every option leads the users to data entry form where the user can key in the data.

Error Message Design

The design of error messages is an important part of the user interface design. As user is bound to commit some errors or other while designing a system the system should be designed to be helpful by providing the user with information regarding the error he/she has committed.

This application must be able to produce output at different modules for different inputs.

Performance Requirements

Performance is measured in terms of the output provided by the application. Requirement specification plays an important part in the analysis of a system. Only when the requirement specifications are properly given, it is possible to design a system, which will fit into required environment. It rests largely in the part of the users of the existing system to give the requirement specifications because they are the people who finally use the system. This is because the requirements have to be known during the initial stages so that the system can be designed according to those requirements. It is very difficult to change the system once it has been designed and on the other hand designing a system, which does not cater to the requirements of the user, is of no use.

The requirement specification for any system can be broadly stated as given below:

- The system should be able to interface with the existing system
- The system should be accurate
- The system should be better than the existing system
- The existing system is completely dependent on the user to perform all the duties.

CHAPTER 9

SOURCE CODE

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

from sklearn.neighbors import KNeighborsClassifier

from sklearn import preprocessing

from sklearn.model_selection import train_test_split

from sklearn import metrics

import seaborn as sns

import matplotlib.pyplot as plt

# definisiin nama coloumn yg berisikan gejala dan diagnosis.

names = ['GP01','GP02','GP03','GP04','GP05', 'GP06','GP07', 'GP08', 'GP09', 'GP10', 'GP11',
'GP12', 'GP13', 'GP14', 'GP15', 'GP16', 'GP17', 'GP18', 'GP19', 'GP20', 'GP21', 'GP22', 'GP23',
'GP24', 'GP25','GP26','GP27','GP28', 'Diagnosis']

# Baca Dataset ke pandas dataframe

Dataset= pd.read_excel('rm1.xls', names = names)

# Data Visualization and Analysis

Dataset['Diagnosis'].value_counts()

# In[27]:

# Count the occurrences of each diagnosis
```

```
diagnosis_counts = Dataset['Diagnosis'].value_counts()

# Create a count plot

plt.figure(figsize=(8, 6))

ax = diagnosis_counts.plot(kind='bar')

plt.xlabel('Diagnosis')

plt.ylabel('Count')

plt.title('Count Plot of Dental Dataset Diagnoses')

for i, count in enumerate(diagnosis_counts):

    ax.text(i, count + 1, str(count), ha='center', va='bottom')

plt.tight_layout()

plt.show()

# In[28]:

Dataset.columns

# In[29]:

Dataset.values

# In[30]:

Dataset.info()

# In[31]:

Dataset.describe(include='all')

# In[32]:

X=Dataset.iloc[:,28]

X.head()
```

```
# In[33]:

y=iris.iloc[:,-1]

# Data Normalization

# In[34]:

X = preprocessing.MinMaxScaler().fit_transform(X) #Min Max Normalization

# Train Test Split

#split data training dan data testing

from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split( X, y, test_size=0.2, random_state= 1)

# Training and Predicting

# In[36]:

knnmodel=KNeighborsClassifier(n_neighbors=6,metric='euclidean')

# In[37]:

knnmodel.fit(X_train,y_train)

# In[38]:

y_predict1=knnmodel.predict(X_test)

# Output Visualization

prediction_output=pd.DataFrame(data=[y_predict1,y_test.values],index=['Predicted
Output','Actual Output'])

prediction_output.iloc[0,:].value_counts()

prediction_output.transpose()

# Accuracy
```

```
from sklearn.metrics import accuracy_score

acc=accuracy_score(y_test,y_predict1)*100

print("KNeighborsClassifier:",acc)

# ### Confusion Matrix

from sklearn.metrics import confusion_matrix

from sklearn.metrics import classification_report

cm=confusion_matrix(y_test.values,y_predict1)

cr=classification_report(y_test.values,y_predict1)

cm1=pd.DataFrame(data=cm,index=['Pulpitis', 'Stomatitis', 'Periodontitis', 'Karies Gigi', 'Abses gigi', 'Gingivitis'],columns=['Pulpitis', 'Stomatitis', 'Periodontitis', 'Karies Gigi', 'Abses gigi', 'Gingivitis'])

# In[48]:

plt.figure(figsize=(8, 6))

sns.heatmap(cm1, annot=True, fmt='d', cmap='Blues')

plt.title('KNeighborsClassifier_Confusion Matrix')

plt.xlabel('Predicted Label')

plt.ylabel('True Label')

plt.show()

# In[49]:

print('\n\nclassification report: \n\n',cr)
```

CHAPTER 10

RESULTS AND DISCUSSION

10.1 Implementation description

The code is a Python script for performing a classification task on a dental dataset using the K-Nearest Neighbors (KNN) algorithm. Below is an explanation of the code's main components and the steps it takes:

- **Import Libraries:** The script begins by importing necessary libraries, including pandas for data manipulation, numpy for numerical operations, matplotlib and seaborn for data visualization, and scikit-learn (sklearn) for machine learning tasks.
- **Define Column Names:** The variable names is defined to specify the column names for the dataset. These column names include features (GP01 to GP28) and the target variable (Diagnosis).
- **Read Dataset:** The dataset is read from an Excel file named 'rml.xls' into a pandas DataFrame called Dataset. The specified column names are applied to the DataFrame.
- **Data Visualization and Analysis:** The code counts the occurrences of each diagnosis category in the "Diagnosis" column and creates a count plot to visualize the distribution of diagnoses.
- **Dataset Information:** The script prints information about the dataset using the info() method, providing details such as the number of non-null entries and data types for each column.
- **Dataset Description:** The describe() method is used to provide summary statistics for the dataset, including count, mean, standard deviation, minimum, and maximum values for numerical columns.

-
- **Data Preparation:** The script extracts the features (X) and the target variable (y) from the dataset. Data normalization is performed on the features using Min-Max scaling to ensure that all features have values within a similar range.
 - **Train-Test Split:** The dataset is split into training and testing sets using the `train_test_split` function. The split ratio is 80% for training and 20% for testing, with a random seed for reproducibility.
 - **K-Nearest Neighbors (KNN) Model:** A KNN classifier model is defined with hyperparameters, such as the number of neighbors (`n_neighbors`) set to 6 and the distance metric (`metric`) set to 'euclidean'. The model is fitted to the training data using `knnmodel.fit(X_train, y_train)`.
 - **Model Prediction:** The KNN model is used to make predictions on the test data using `knnmodel.predict(X_test)`.
 - **Output Visualization:** The script creates a DataFrame `prediction_output` to display the predicted and actual outputs. The counts of predicted categories are shown.
 - **Accuracy Calculation:** The script calculates the accuracy of the KNN model on the test data using `accuracy_score` from scikit-learn.
 - **Confusion Matrix:** A confusion matrix and a classification report are generated to evaluate the model's performance. The confusion matrix shows the counts of true positive, true negative, false positive, and false negative predictions. The classification report provides additional metrics, including precision, recall, and F1-score for each class.
 - **Confusion Matrix Visualization:** The confusion matrix is visualized using a heatmap, making it easier to interpret the model's performance on different classes.
 - **Print Classification Report:** The classification report is printed to provide detailed performance metrics for each class.

10.2 Dataset description

The dataset contains following columns

GP01: This column likely represents a data point or measurement, but its specific meaning depends on the context of the dataset.

GP02: Similar to GP01, GP02 is another data point or measurement whose interpretation depends on the context.

GP03: This column may also represent a data point or measurement, but without context, it's unclear what it measures.

GP04 through GP28: These columns likely represent various data points, measurements, or features. The interpretation of each column depends on the specific data and context within your dataset.

Diagnosis: This column likely contains the diagnosis or classification labels for each data entry. It could be binary (e.g., yes/no, 0/1) or multi-class (with multiple possible diagnosis labels). The "Diagnosis" column is typically the target variable in many machine learning or statistical analyses.

10.3 Results and description

Figure 1 displays a portion of the dataset used for dental diagnosis. It might show a few rows and columns to give an overview of the data, providing a glimpse of how the information is structured. Figure 2 is a graphical representation, possibly a bar plot or histogram, that visualizes the distribution of classes in the target column of the dataset. It shows how many samples belong to each diagnosis category, providing insights into the class distribution. Figure 3 display summary statistics of the dataset. This includes measures used for dental diagnosis.

	GP01	GP02	GP03	GP04	GP05	GP06	GP07	GP08	GP09	GP10	...	GP20	GP21	GP22	GP23	GP24	GP25	GP26	GP27	GP28	Diagnosis
0	0.5	0.75	0.5	0.00	0.0	0.0	0.0	0.75	0.50	0.00	...	0.0	0.00	0.00	0.00	0.0	0.0	0.00	0.0	0.00	Gingivitis
1	0.0	0.00	0.0	0.75	0.0	1.0	0.0	0.50	0.75	0.00	...	0.0	0.00	0.00	0.00	0.0	0.0	0.00	0.0	0.00	Gingivitis
2	0.0	0.00	0.0	0.00	0.0	0.0	0.0	0.00	0.50	0.75	...	0.0	0.75	0.00	1.00	0.0	0.0	0.00	0.0	0.00	Karies Gigi
3	0.0	0.75	0.0	0.00	0.5	0.0	0.0	0.00	0.00	0.00	...	0.0	0.00	0.00	0.00	0.0	0.0	0.00	1.0	0.00	Periodontitis
4	0.0	0.00	0.0	0.00	0.0	0.0	0.0	1.00	0.25	0.00	...	0.0	0.00	0.00	0.75	0.0	0.0	0.00	0.0	0.00	Karies Gigi
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
95	0.0	0.00	0.0	0.00	0.0	0.0	0.0	0.00	0.00	0.00	...	0.0	0.00	0.25	0.00	0.0	0.0	0.00	0.0	0.00	Pulpitis
96	0.5	1.00	0.0	0.75	0.0	1.0	0.0	0.00	0.00	0.00	...	0.0	0.00	0.00	0.00	0.0	0.0	0.00	1.0	0.00	Periodontitis
97	0.0	0.00	0.0	0.75	1.0	1.0	0.0	0.00	0.00	0.00	...	0.0	0.00	0.00	0.00	0.5	0.0	0.25	0.0	0.00	Abses gigi
98	0.0	0.00	0.0	0.00	0.0	0.0	0.0	0.00	0.75	0.75	...	0.0	0.00	0.00	0.00	0.0	0.0	0.00	0.0	0.00	Karies Gigi
99	0.0	1.00	0.0	0.75	0.0	1.0	0.5	0.00	0.00	0.00	...	0.0	0.00	0.00	0.00	0.0	0.0	0.00	0.0	0.75	Gingivitis

100 rows × 29 columns

Figure 1: sample dataset used for dental diagnosis

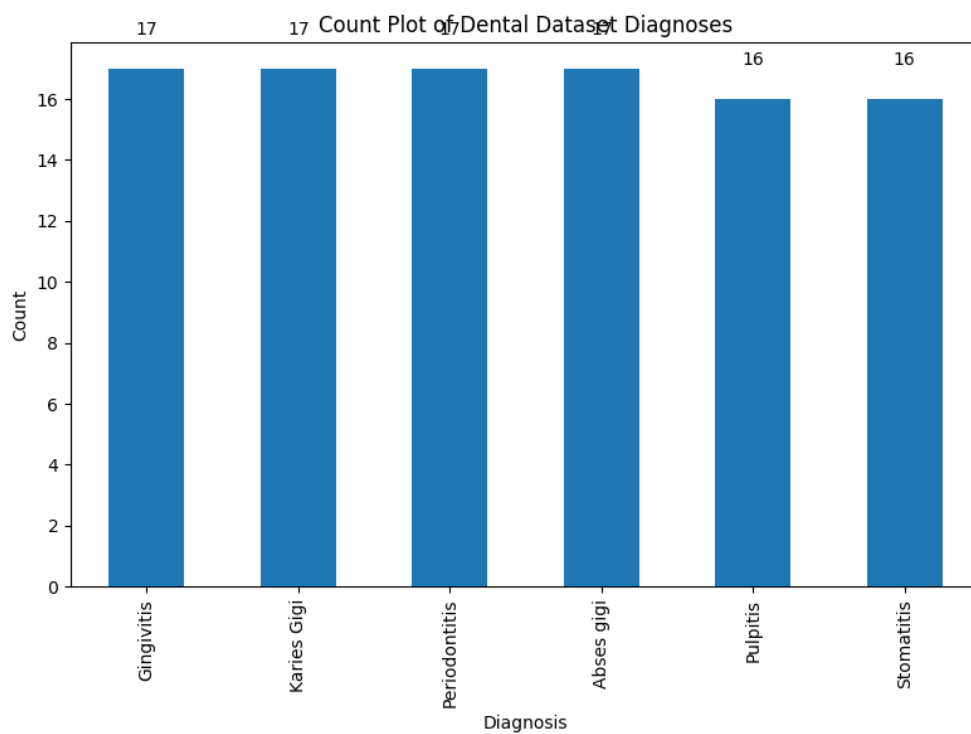


Figure 2: count plot of target column of dataset

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 29 columns):
#   Column      Non-Null Count  Dtype
---  -
0   GP01         100 non-null    float64
1   GP02         100 non-null    float64
2   GP03         100 non-null    float64
3   GP04         100 non-null    float64
4   GP05         100 non-null    float64
5   GP06         100 non-null    float64
6   GP07         100 non-null    float64
7   GP08         100 non-null    float64
8   GP09         100 non-null    float64
9   GP10         100 non-null    float64
10  GP11         100 non-null    float64
11  GP12         100 non-null    float64
12  GP13         100 non-null    float64
13  GP14         100 non-null    float64
14  GP15         100 non-null    float64
15  GP16         100 non-null    float64
16  GP17         100 non-null    float64
17  GP18         100 non-null    float64
18  GP19         100 non-null    float64
19  GP20         100 non-null    float64
20  GP21         100 non-null    float64
21  GP22         100 non-null    float64
22  GP23         100 non-null    float64
23  GP24         100 non-null    float64
24  GP25         100 non-null    float64
25  GP26         100 non-null    float64
26  GP27         100 non-null    float64
27  GP28         100 non-null    float64
28  Diagnosis    100 non-null    object
dtypes: float64(28), object(1)
memory usage: 22.8+ KB

```

Figure 3: summary of a dataset used for dental diagnosis

Figure 4 showcase specific features (columns) of the dataset that are relevant to dental diagnosis. It display information about individual attributes, potentially providing details about their nature and values. Figure 5 focused view of the target column. It could display the unique categories or labels present in the target column, providing a clear understanding of the different diagnoses or classes.

	GP01	GP02	GP03	GP04	GP05	GP06	GP07	GP08	GP09	GP10	...	GP19	GP20	GP21	GP22	GP23	GP24	GP25	GP26	GP27	GP28
0	0.5	0.75	0.5	0.00	0.0	0.0	0.0	0.75	0.50	0.00	...	0.0	0.0	0.00	0.00	0.00	0.0	0.0	0.00	0.0	0.00
1	0.0	0.00	0.0	0.75	0.0	1.0	0.0	0.50	0.75	0.00	...	0.0	0.0	0.00	0.00	0.00	0.0	0.0	0.00	0.0	0.00
2	0.0	0.00	0.0	0.00	0.0	0.0	0.0	0.00	0.50	0.75	...	0.0	0.0	0.75	0.00	1.00	0.0	0.0	0.00	0.0	0.00
3	0.0	0.75	0.0	0.00	0.5	0.0	0.0	0.00	0.00	0.00	...	0.0	0.0	0.00	0.00	0.00	0.0	0.0	0.00	1.0	0.00
4	0.0	0.00	0.0	0.00	0.0	0.0	0.0	1.00	0.25	0.00	...	0.0	0.0	0.00	0.00	0.75	0.0	0.0	0.00	0.0	0.00
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
95	0.0	0.00	0.0	0.00	0.0	0.0	0.0	0.00	0.00	0.00	...	0.0	0.0	0.00	0.25	0.00	0.0	0.0	0.00	0.0	0.00
96	0.5	1.00	0.0	0.75	0.0	1.0	0.0	0.00	0.00	0.00	...	0.0	0.0	0.00	0.00	0.00	0.0	0.0	0.00	1.0	0.00
97	0.0	0.00	0.0	0.75	1.0	1.0	0.0	0.00	0.00	0.00	...	0.0	0.0	0.00	0.00	0.00	0.5	0.0	0.25	0.0	0.00
98	0.0	0.00	0.0	0.00	0.0	0.0	0.0	0.00	0.75	0.75	...	0.0	0.0	0.00	0.00	0.00	0.0	0.0	0.00	0.0	0.00
99	0.0	1.00	0.0	0.75	0.0	1.0	0.5	0.00	0.00	0.00	...	0.0	0.0	0.00	0.00	0.00	0.0	0.0	0.00	0.0	0.75

100 rows × 28 columns

Figure 4: feature of a data frame used for dental diagnosis

```
0      Gingivitis
1      Gingivitis
2      Karies Gigi
3      Periodontitis
4      Karies Gigi
...
95     Pulpitis
96     Periodontitis
97     Abses gigi
98     Karies Gigi
99     Gingivitis
Name: Diagnosis, Length: 100, dtype: object
```

Figure 5: Target column of a dataset used for dental diagnosis

```
array([[0.5 , 0.75, 0.5 , ..., 0. , 0. , 0. ],
       [0. , 0. , 0. , ..., 0. , 0. , 0. ],
       [0. , 0. , 0. , ..., 0. , 0. , 0. ],
       ...,
       [0. , 0. , 0. , ..., 0.25, 0. , 0. ],
       [0. , 0. , 0. , ..., 0. , 0. , 0. ],
       [0. , 1. , 0. , ..., 0. , 0. , 1. ]])
```

Figure 6: features of dataset after preprocessing

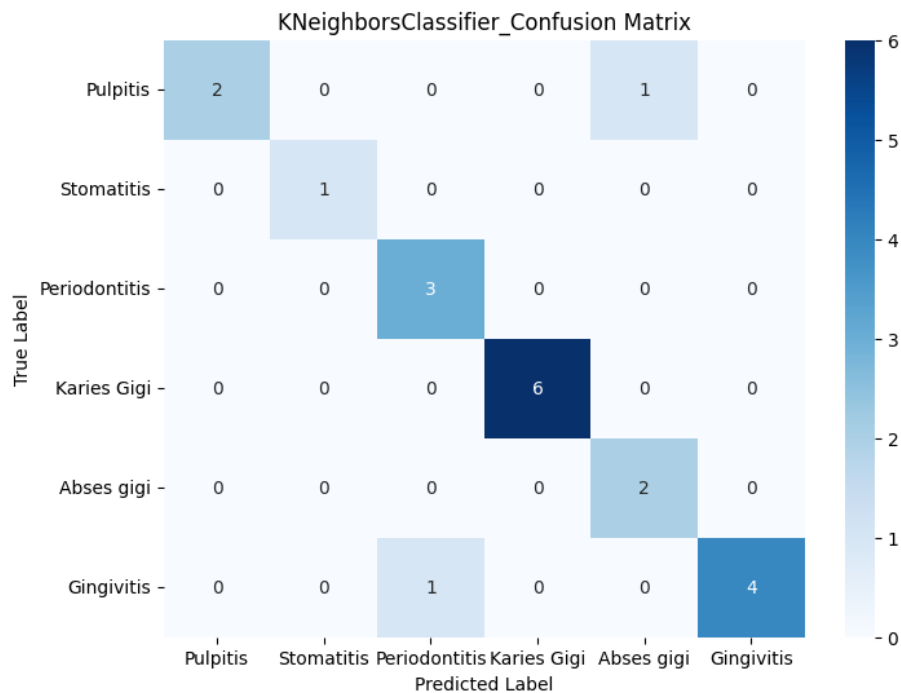


Figure 7: confusion matrix of KNN Classifier

Figure 6 shows the dataset after some form of preprocessing. It could include information about transformed or engineered features that are used for dental diagnosis. Figure 7 is a representation of the confusion matrix generated by a K-Nearest Neighbors (KNN) classifier. The confusion matrix provides information about the model's performance, showing how many samples were correctly or incorrectly classified for each class. Figure 8 presents a classification report for the KNN classifier. A classification report provides various evaluation metrics (such as precision, recall, F1-score, and support) for each class, offering a comprehensive assessment of the model's performance.

classification report:

	precision	recall	f1-score	support
Abses gigi	1.00	0.67	0.80	3
Gingivitis	1.00	1.00	1.00	1
Karies Gigi	0.75	1.00	0.86	3
Periodontitis	1.00	1.00	1.00	6
Pulpitis	0.67	1.00	0.80	2
Stomatitis	1.00	0.80	0.89	5
accuracy			0.90	20
macro avg	0.90	0.91	0.89	20
weighted avg	0.93	0.90	0.90	20

Figure 8: classification report of KNN Classifier

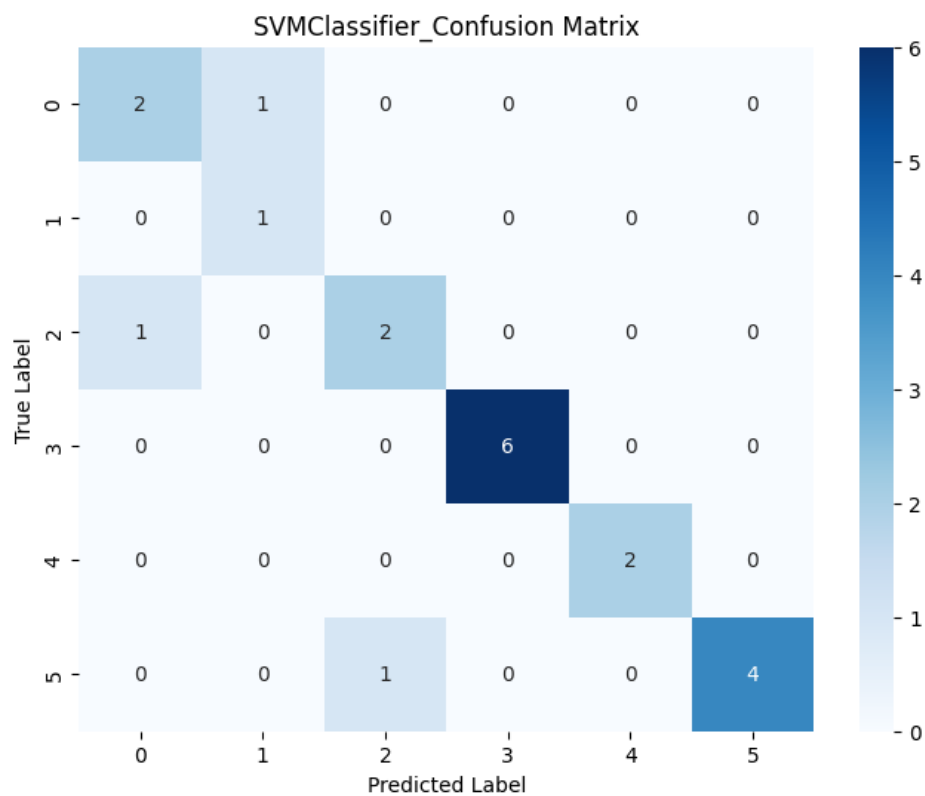


Figure 9: confusion matrix of svm classifier

SVM Classifier Classification_report:				
	precision	recall	f1-score	support
Abses gigi	0.67	0.67	0.67	3
Gingivitis	0.50	1.00	0.67	1
Karies Gigi	0.67	0.67	0.67	3
Periodontitis	1.00	1.00	1.00	6
Pulpitis	1.00	1.00	1.00	2
Stomatitis	1.00	0.80	0.89	5
accuracy			0.85	20
macro avg	0.81	0.86	0.81	20
weighted avg	0.88	0.85	0.86	20

Figure 10: classification report of svm classifier

Figure 9 is the visual representation provides a detailed breakdown of the SVM model's performance. It shows how many samples were correctly and incorrectly classified for each class. The diagonal elements represent correct predictions, while off-diagonal elements represent misclassifications. Figure 10 is a report provides a comprehensive evaluation of the SVM model's performance. It includes metrics such as precision, recall, F1-score, and support for each class. Precision indicates the accuracy of positive predictions, while recall measures the model's ability to identify all positive instances. F1-score is the harmonic mean of precision and recall, providing a balanced assessment. Support refers to the number of occurrences of each class in the dataset.

CHAPTER 11

CONCLUSION AND FUTURE SCOPE

11. 1 Conclusion

In conclusion, the development of an expert system for enhanced dental disease diagnosis represents a significant step forward in improving the accuracy and efficiency of dental healthcare. By harnessing the capabilities of artificial intelligence, this system has the potential to augment the skills and expertise of dentists, leading to more precise diagnoses, better treatment plans, and ultimately, improved patient outcomes. It addresses the limitations of traditional manual diagnosis, such as human errors and biases, while also providing access to real-time updates from dental research, ensuring that dental professionals stay up-to-date with the latest advancements in the field. This project underscores the transformative role of AI in healthcare and sets the stage for further innovation in medical diagnostics.

11.2 Future Scope

Looking to the future, the scope for this application is promising. Firstly, continuous refinement and expansion of the system's knowledge base will be essential to keep pace with the rapidly evolving field of dentistry. Integration with emerging technologies, such as 3D imaging and genomic data analysis, could further enhance the system's diagnostic capabilities. Additionally, the application could be adapted for teledentistry, enabling remote consultations and expanding access to dental care, particularly in underserved areas. Collaboration with dental professionals and institutions for data sharing and validation will be crucial for the system's ongoing improvement. Moreover, the principles and techniques developed in this project could serve as a blueprint for similar expert systems in other medical specialties, ushering in a new era of AI-powered healthcare diagnostics. Ultimately, the future scope involves not only the continual advancement of this specific application but also its potential to revolutionize the broader landscape of medical diagnosis and treatment.

CHAPTER 12

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Project Details				
Academic Year		2023-2024		
Title of the Project		Expert System Application Design For Enhanced Dental Disease Diagnosis		
Name of the Students and Hall Ticket No.		SANEM SREE MALIK GOUD (20RA1A0586) S.SAI KIRAN (21RA5A0513) N.SWETHA (20RA1A0594)		
Name of the Guide		Mr. T .Ranindar		
Project PO Mapping				
Name of Course from which Principles are applied in this Project	Related Course Outcome Number	Description of the application	Page Number	Attained
Python Programming Software Engineering (C313)	C313.1	Students described the basis for their problem statement.	01	PO2
Machine Learning, Python Programming, Data Mining (C322, C411)	C322.2, C411.2	Students explained about dental Disease diagnosis using ML with python programming	01	PO1
Software Engineering, Python Programming (C313)	C313.3	Students identified the existingsystem and its Drawbacks and proposed a Solution to it.	16-24	P02,P03
Software Engineering (C313)	C313.1	Students identified the hardware andsoftware required for the project	46	PO5
Design Patterns, Software Engineering, (C313, C222)	C313.2, C222.3,	Students explain the flow of the project using UML diagrams.	25-28	P03,P05, PO9, PSO3

Python Programming, Data Mining (C411)	C411.2	Students explained about python programming language and decided the packages for the solution of the problem.	29-45	PO2, PO3,PO4
Python Programming		Students Developed code for the problem Statement.	52-55	PO3,PO4,PO5
Future Scope		Students explained about how they would like to take their project further and develop it as their future scope	65	PO12,PS02
Bibliography		Listed the references from which the literature was collected	66-68	PO8,PO12
ENG		Prepared the thesis and intermediate progress reports and explained to the review panel. Also, continuously caught up with guide and explained the progress.		PO9,PO10

SIGNATURE OF STUDENT

SIGNATURE OF INTERNAL
GUIDE